Exercise J1

Q1. If y =$\log(x+\sqrt{x^2+a^2})$ then dy/dx equals–

(A) $\frac{x}{\sqrt{x^2+a^2}}$ (B) $\frac{1}{\sqrt{x^2+a^2}}$ (C) $\frac{-1}{\sqrt{x^2+a^2}}$ (D) $\frac{1}{\sqrt{x^2-a^2}}$

Ans. B

Sol. $y = \log\left(x + \sqrt{x^2 + a^2}\right)$

$\frac{dy}{dx} = \left(\frac{1}{x+\sqrt{x^2+a^2}}\right)\left(1+\frac{x}{x+\sqrt{x^2+a^2}}\right)$

$= \frac{1}{\sqrt{x^2 + a^2}}$

Q2. If y = 1+$\frac{x^2}{2!}+\frac{x^3}{3!}$+ .....then dy/dx equals–

(A) y – x – 1 (B) y + x – 1 (C) y + x (D) y – x

Ans. B

Sol. $y = +1+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+ \ldots\ldots\ldots\ldots$

$\frac{dy}{dx} = x+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+ \ldots\ldots\ldots\ldots$

= x + y – 1

Q3. The derivative of tan x° is –

(A) $\left(\frac{\pi}{180}\right)\cos^2 x$ (B) $\left(\frac{\pi}{180}\right)\sec^2 x$ (C) $\left(\frac{\pi}{180}\right)\cos^2 x^\circ$ (D) $\left(\frac{\pi}{180}\right)\sec^2 x^\circ$

Ans. D

Sol. y = tan (x°)

$y = \tan\left(\frac{\pi x}{180}\right)$

$\frac{dy}{dx} = sec^2\left(\frac{\pi x}{180}\right)\times\frac{\pi}{180}$

$= \frac{\pi}{180^\circ}\sec^2(x)$

Q4. If $x^y + y^x = a^b$ then $\frac{dy}{dx}$ equals–

(A) $\frac{yx^{y-1}+y^x\log y}{x^y\log x+xy^{x-1}}$ (B) $\frac{x^y\log x+x.y^{x-1}}{y.x^{y-1}+\log y+y^x}$ (C) $-\frac{yx^{y-1}+y^x\log y}{x^y\log x+xy^{x-1}}$

(D) None of these

Ans. C

Sol. $x^y + y^x = a^b$

diff. wrt x

$$x^y\left(\frac{y}{x} + ln\ x + \frac{dy}{dx}\right) + y^x\left(ln\ y + \frac{x}{y}\frac{dy}{dx}\right) = 0$$

$$\left(x^y\ ln\ x + y^{x-1}x\right)\frac{dy}{dx} = -\left(y^x ln\ y + yx^{y-1}\right)$$

$$\frac{dy}{dx} = -\left(\frac{y^x\ ln\ y + yx^{y^{-x}}}{x^y\ ln\ x + xy^{x^{-1}}}\right)$$

Q5. If sin y = x sin (a+y) then dy/dx is–

(A) $\frac{\sin(a+y)}{\sin^2 a}$ (B) $\frac{\sin^2(a+y)}{\sin a}$ (C) $\frac{\sin(a+y)}{\sin^2 a}$ (D) None of these

Ans. B

Sol. sin y = x sin (a + y)

$$x = \frac{sin\ y}{sin\ (a+y)}$$

$$\frac{dy}{dx} = \frac{sin(a+y)\ cos\ y - siny\ cos(a+y)}{sin^2\ (a+y)}$$

$$\frac{dx}{dy} = \frac{sin(a)}{sin^2(a+y)}$$

$$\frac{dy}{dx} = \frac{sin^2\ (a+y)}{sin\ a}$$

Q6. If $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$, then dy/dx equals-

(A) -1 (B) 1 (C) x (D) $\sqrt{x}$

Ans. A

Sol. $y = (1 - x^{1/4})\ (1 + x^{1/4})$

$= (1 - x^{1/2})\ (1 + x^{1/2})$

$y = 1 - x$

$$\frac{dy}{dx} = -1$$

Q7. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then $\frac{dy}{dx}$ equals-

(A) $\frac{1}{(1+x)^2}$ (B) $-\frac{1}{(1+x)^2}$ (C) $\frac{1}{1+x^2}$ (D) None of these

Ans. B

Sol. $x^2(1 + y) = y^2(1 + x) \Rightarrow x^2 - y^2 + xy(x - y) = 0$

$\Rightarrow (x - y)(x + y + xy) = 0$

Now $x \neq y$ [does not satisfy the given equation]

$\therefore x + y + xy = 0 \Rightarrow y = \frac{-x}{1+x}$

$\therefore \frac{dy}{dx} = \frac{-(1 + x) + x}{(1+x)^2} = \frac{1}{(1+x)^2}$

Q8. If $y = \sin^{-1}\sqrt{\sin x}$, then $\frac{dy}{dx}$ equals-

(A) $\frac{2\sqrt{\sin x}}{\sqrt{1+\sin x}}$ (B) $\frac{\sqrt{\sin x}}{\sqrt{1-\sin x}}$ (C) $\frac{1}{2}\sqrt{1+\cos ec\, x}$ (D) $\frac{1}{2}\sqrt{1-\cos ec\, x}$

Ans. C

Sol. $y = \sin^{-1}\sqrt{sin\ x}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1 - sin\ x}} \times \frac{cos\ x}{2\sqrt{sin\ x}} = \frac{\sqrt{1 + sin\ x}}{2\sqrt{sin\ x}}$

$= \frac{1}{2}\sqrt{cosec\ x + 1}$

Q9. If $x^y y^x = 1$, then $\frac{dy}{dx}$ equals-

(A) $\frac{x(y + x\log y)}{y(x + y\log x)}$ (B) $-\frac{x(x + y\log y)}{y(y + x\log x)}$ (C) $\frac{y(y + x\log y)}{x(x + y\log x)}$ (D) $-\frac{y(y + x\log y)}{x(x + y\log x)}$

Ans. D

Sol. $x^y y^y = 1$

take logarithm

$y \log x + x \log y = 0$

diff. wrt x

$\frac{y}{x} + (\log x)\frac{dy}{dx} + \log y + \frac{dy}{dx} = 0$

$\left(log\ x + \frac{dy}{dx}\right)\frac{dy}{dx} = -\left(log\ y + \frac{y}{x}\right)$

$\left(\frac{x + y\cos x}{y}\right)\frac{dy}{dx} = -\left(\frac{x\ log\ y + y}{x}\right)$

$\frac{dy}{dx} = -\frac{y(x\ log\ y + y)}{x(x + y\ log\ x)}$

Q10. If $y = \tan{-1}\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}$, then $\frac{dy}{dx}$ equals-

(A) $-\frac{1}{2\sqrt{1-x^2}}$ (B) $-\frac{1}{\sqrt{1-x^4}}$ (C) $-\frac{x}{\sqrt{1-x^4}}$ (D) $-\frac{x}{2\sqrt{1-x^4}}$

Ans. C

Sol. y = tan $\dfrac{\sqrt{1 + x}^2 + \sqrt{1 - x}^2}{\sqrt{1 + x}^2 - \sqrt{1 - x}^2}$

Put $x^2 = \cos 2\theta$

$= \tan^{-1}\left(\dfrac{\cos\theta + \sin\theta}{\cos\theta - \cos\theta}\right)$

$= \tan^{-1}\left(\dfrac{1 + \tan\theta}{1 - \tan\theta}\right)$

$= \tan^{-1}(1) + \tan^{-1}(h, \theta)$

$= \dfrac{\pi}{4} + \theta = \dfrac{\pi}{4} + \dfrac{1}{2}\cos^{-1}(x^2)$

$\dfrac{dy}{dx} = \dfrac{1}{2}\left(\dfrac{-1}{\sqrt{1 - x^4}}\right)(2x)$

$\dfrac{dy}{dx} = \dfrac{-x}{\sqrt{1 - x^4}}$

Q11. If $f(x) = \log_x(\text{In } x)$, then at x = e, f'(x) equals-

(A) 0 (B) 1 (C) e (D) 1/e

Ans. D

Sol. (x) = logx(*l*nx)

$\Rightarrow \quad f(x) = \dfrac{ln(lnx)}{lnx}$

$\Rightarrow \quad f'(x) = \dfrac{lnx\left(\frac{1}{lnx}\cdot\frac{1}{x}\right) - \frac{1}{x}(lnx)}{(lnx)^2}$

$\Rightarrow \quad f'(e) = \dfrac{\frac{1}{e} - 0}{1} = 1/e$

Q12. If $y = e^3x \sin 4x$, then the value of dy/dx is-

(A) $e^{3x}\sin\left(4x + \tan^{-1}\dfrac{4}{3}\right)$ (B) $e^{3x}\cos\left(4x + \tan^{-1}\dfrac{4}{3}\right)$ (C) $5e^{3x}\sin\left(4x + \tan^{-1}\dfrac{4}{3}\right)$

(D) $5e^{3x}\cos\left(4x + \tan^{-1}\dfrac{4}{3}\right)$

Ans. C

Sol. $y = e^{3x}\sin 4x$

$\dfrac{dy}{dx} = e^{3x}\sin 4x + 4e^{3x}\cos 4x$

$= e^{3x}(3\sin 4x + 4\cos 4x)$

$= 5e^{3x}\left(\dfrac{3}{5}\sin 4x + \dfrac{4}{5}\cos\right)$

$= 5e^{3x} \sin(4x + \phi)$ when $\cos = \phi \frac{3}{5}$

$$\sin\phi = \frac{4}{5}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$$

Q13. If $x + y = x^y$, then dy/dx equals-

(A) $\frac{yx^{y-1} - 1}{1 - x^y \log x}$ (B) $\frac{yx^{y-1} - 1}{x^y \log x - 1}$ (C) $\frac{yx^{y-1} + 1}{x^y \log x + 1}$ (D) None of these

Ans. A

Sol. $x + y = x^y$

Diff. wrt x

$$1 + \frac{dy}{dx} = x^y \left(\frac{y}{x} + lnx\frac{dy}{dx}\right)$$

$$\frac{dy}{dx}(1 - x^y \ln x) = yx^{y-1} - 1$$

$$\frac{dy}{dx} = \frac{yx^{x-1} - 1}{1 - x^y\, lnx}$$

Q14. The derivative of x^{a^x} is-

(A) $x^{a^x}[a^x x - 1 + a^x \log a \log x]$ (B) $x^{a^x}[xa^x + a^x \log x]$ (C) $x^{a^x}[a^x + xa^x \log x]$

(D) None of these

Ans. A

Sol. $y = x^{a^x}$

Take logarithm

$\ln y = a^x \ln x$

Diff. wrt x

$$\frac{1}{y}\frac{dy}{dx} = \frac{a^x}{x} + (\ln x)\, a^x \ln a$$

$$\frac{dy}{dx} = y\left(\frac{a^x}{x} + a^x(lnx)(lna)\right)$$

$$\frac{dy}{dx} = x^{a^x}\left(\frac{a^x}{x} + a^x(lnx)(lna)\right)$$

Q15. If $(\cos x)^y = (\sin y)^x$, then dy/dx equals-

(A) $\frac{\log\sin y - y\tan x}{\log\cos x + x\cot y}$ (B) $\frac{\log\sin y + y\tan x}{\log\cos x - x\cot y}$ (C) $\frac{\log\sin y + y\tan x}{\log\cos x + x\cot y}$

(D) None of these

Ans. B

Sol. $(\cos x)^y = (\sin y)^x$

Take logarithm

y ln cos x = x lin sin y

Diff. wrt

$\left(ln(cosx)\right)\frac{dy}{dx}$ = y tan = ln (sin y) + (cot y) x $\frac{dy}{dx}$

(ln cos x – x cot y) $\frac{dy}{dx}$ = ln (sin y) + y $\tan^{-1}$

$$\frac{dy}{dx} = \frac{ln\ (sin\ y)\ +\ y\ tan\ x}{ln\ (cos\ x) -\ x\ cot\ y}$$

Q16. If y = $\sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}]$, then dy/dx can be equal to–

(A) $\frac{2}{1+x^2}$ (B) $\frac{1}{1+x^2}$ (C) $\frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x}\sqrt{1-x}}$ (D) None of these

Ans. C

Sol. y = $\sin^{-1}$ (x) – $\sin^{-1}\sqrt{x}$

$$\frac{dy}{ln} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x}\sqrt{1-x^2}}$$

Q17. The differential coefficient (x > 0) of $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ w.r.t $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ can be equal to –

(A) 1/2 (B) – 1/2 (C) 1 (D) –1

Ans. A

Sol. y = $\sin^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ z = $\cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)$

y = $\tan^{-1}$ (x) z = 2 $\tan^{-1}$ (x)

$$\frac{dy}{lt} = \frac{\frac{cos}{dx}}{\frac{dt}{dx}} = \frac{\frac{1}{1+x^2}}{\frac{2}{2(1+x)}} = \frac{1}{2}$$

Q18. If y = $\sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\sqrt{x}\ldots\ldots\infty}}}$, then the value of dy/dx is-

(A) $\frac{xy^2}{2-y\log x}$ (B) $\frac{x^2}{y(2-y\log x)}$ (C) $\frac{y^2}{x(2-y\log x)}$ (D) $\frac{y^2}{x(2+y\log x)}$

Ans. C

Sol. y = $(\sqrt{x})^y$

Take logarithm

ln y = y ln $\sqrt{x}$

ln y = $\frac{y}{2}$ ln x

Diff wrt x

$\frac{1}{y}\ \frac{dy}{dx} = \frac{y}{2\mu} + \frac{ln\mu}{2}\frac{dy}{dx}$

$\frac{dy}{dx}\left(\frac{2}{y} - ln\ x\right) = \frac{y}{x}$

$\left(\frac{2 - y\ ln\ x}{y}\right) = \frac{y}{x}$

$\frac{dy}{dx} = \frac{y^2}{x(x - y\ ln\ x)}$

Q19. If y = (tan x)$^{\tan x .... \infty}$, then tan x $\left(\frac{dy}{dx}\right)$equals-

(A) $\frac{y^2 \sec^2 x}{1 - y \log \tan x}$ (B) $\frac{y^2 \sec^2 x}{1 + y \log \tan x}$ (C) $\frac{y \sec^2 x}{1 - y \log \tan x}$ (D) None of these

Ans. A

Sol. y = (tan x)^y

Take logarithm

log y = y log tan x

Diff. wrt x

$\frac{1}{y}\ \frac{dy}{dx} = \frac{y\ sec^2\ x}{tan\ x}$+ (log tan x) $\frac{dy}{dx}$

$\frac{dy}{dx}\left(\frac{1 - y\ log\ tan\ x}{y}\right) = \frac{y\ sec^2\ x}{tan\ x}$(tan x) $\left(\frac{dy}{dx}\right)\frac{y^2\ \ sec^2\ x}{1 - y\ log\ tan\ x}$

Q20. If y = $\frac{x}{a + \frac{x}{b + \frac{x}{a + \frac{x}{b + ...}}}}$, then dy/dx equals–

(A) $\frac{a}{a(b + 2y)}$ (B) $\frac{b}{b + 2y}$ (C) $\frac{a}{b(b + 2y)}$ (D) None of these

Ans. A

Sol. y = $\frac{x}{a + \frac{x}{b + y}}$

ay + $\frac{xy}{b\ +\ y}$= x

ay = x $\left(1 - \frac{y}{b\ +\ y}\right)$

$ay = \frac{xy}{b + y}$

$xb = ay\,(b + y)$

$x = \frac{ay(b + y)}{b}$

Diff. wrt y

$\frac{dx}{dy} = \frac{a}{b}\,(b + 2y)$

$\frac{dy}{dx} = \frac{b}{a(b + 2y)}$

Q21. If $y = \log_y x$, then $= \frac{dy}{dx}$

(A) $\frac{1}{x + \log y}$ (B) $\frac{1}{\log x(1 + y)}$ (C) $\frac{1}{x(1 + \log y)}$ (D) $\frac{1}{y + \log x}$

Ans. C

Sol. $y = \frac{log\ x}{log\ y}$

$y \log y = \log x$

Diff. wrt x

$(1 + \log y)\,\frac{dy}{dx} = \frac{1}{x}$

$\frac{dy}{dx} = \frac{1}{x(1 + log\ y)}$

Q22. If $x = 3 \cos \theta - 2 \cos^3 \theta$ and $y = 3 \sin \theta - 2 \sin^3 \theta$, then $\frac{dy}{dx} =$

(A) $\sin\theta$ (B) $\cos\theta$ (C) $\tan\theta$ (D) $\cot\theta$

Ans. D

Sol. $x = 3\cos \theta - 2\cos^3 \theta$

$\frac{dy}{dx} = -3 \sin \theta - 1\ 6 \cos^2 \theta \sin \theta$

$= 3 \sin \theta\,(-1 + 2 \cos^2 \theta)$

$= 3 \sin \theta \cos^2 \theta$

$y = 3\sin \theta - 2\sin^3 \theta$

$\frac{dy}{dx} = -3 \cos \theta - 1\ 6 \sin^2 \theta \cos \theta$

$= 3 \cos \theta\,(1 - 2 \sin^2 \theta)$

$= 3 \cos \theta \cos^2 \theta$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \cot\theta$$

Q23. Let f(x) be a polynomial function of second degree. If f(1) = f(– 1) and a, b, c are in A.P. then f' (a), f'(b) and f'(c) are in-

(A) Arithmetic - Geometric Progression (B) A.P. (C) G.P. (D) H.P.

Ans. B

Sol. f(x) is a polynomial function

$f(x) = ax^2 + bx + c \qquad f(1) = f(-1)$

$a + b + c = a - b + c \qquad b = 0$

a, b, c in A.P. $\quad b = \frac{a+c}{2} \quad a = -c$

$f(x) = ax^2 + bx + c \qquad f'(x) = 2ax + b$

$f'(a) = 2a^2 + b \qquad f'(c) = 2ac + b$

$f'(b) = 2ab + b$ then f'(a), f'(b), f'(c)

$f'(b) = 0 \quad f'(a) = 2a^2 \qquad f'(c) = -2a^2$

so that f' (a), f'(b), f' (c) are in A.P.

Q24. If $x = e^{y+e^{y+.....to\ \infty}}$, x > 0, then $\frac{dy}{dx}$ is

(A) $\frac{x}{1+x}$ (B) $\frac{1}{x}$ (C) $\frac{1-x}{x}$ (D) $\frac{1+x}{x}$

Ans. C

Sol. $x = e^{y + e^{y+......\infty}} \qquad x > 0 \qquad \frac{dy}{dx} = ?$

$x = e^{y+x} \qquad 1 = e^{y+x}\left(1 + \frac{dy}{dx}\right)$

$\frac{1}{x} = 1 + \frac{dy}{dx} \qquad \frac{1-x}{x} = \frac{dy}{dx}$

Q25. If $x^m . y^n = (x + y)^{m+n}$, then $\frac{dy}{dx}$ is

(A) $\frac{x+y}{xy}$ (B) xy (C) $\frac{x}{y}$ (D) $\frac{y}{x}$

Ans. D

Sol. $x^m y^n = (x + y)^{m+n}$

taking log both sides

$m\, ln\, x + n\, ln\, y = (m + n) . ln\, (x + y)$

$$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \left(\frac{(m+n)}{x+y}\right)\left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = \left(\frac{n}{y} - \frac{(m+n)}{x+y}\right) - \frac{m\ +\ n}{x\ +y} - \frac{m}{x}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y}{x}$$

Q26. y= $\frac{1}{1+(\tan\theta)^{\sin\theta-\cos\theta}+(\cot\theta)^{\cos\theta-\cot\theta}} + \frac{1}{1+(\tan\theta)^{\cos\theta-\sin\theta}+(\sin\theta)^{\sin\theta-\cot\theta}} + \frac{1}{1+(\tan\theta)^{\cos\theta-\cot\theta}+(\cot\theta)^{\cot\theta-\sin\theta}}$ then $\frac{dy}{d\theta}$ at $\theta = \pi/3$ is :

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) None of these

Ans. A

Sol. y = a + b + c

$$a = \frac{1}{1+(tan\ \theta)^{sin\ \theta\ -\ cos\ \theta} +\ (tan\ \theta)^{cot\ \theta}}$$

$$a = \frac{(tan\ \theta)^{cos\ \theta}}{(tan\ \theta)^{cos\ \theta} + (tan\ \theta)^{sin\ \theta} + (tan\ \theta)^{cot\ \theta}}$$

$$b = \frac{1}{1+(tan\ \theta)^{cos\ \theta\ -\ sin\ \theta} +\ (tan\ \theta)^{cot\ \theta\ - sin\ \theta}}$$

$$b = \frac{(tan\ \theta)^{sin\ \theta}}{(tan\ \theta)^{sin\ \theta} + (tan\ \theta)^{cos\ \theta} + (tan\ \theta)^{cot\ \theta}}$$

$$c = \frac{1}{1+(tan\ \theta)^{cos\ \theta\ -\ cot\ \theta} +\ (tan\ \theta)^{sin\ \theta\ - cot\ \theta}}$$

$$c = \frac{(tan\ \theta)^{cot\ \theta}}{(tan\ \theta)^{cot\ \theta} + (tan\ \theta)^{cos\ \theta} + (tan\ \theta)^{sin\ \theta}}$$

So y = a + b + c

y = – 1

$$\frac{dy}{dx} = 0$$

Q27. Let f '(x) = sin(x^2) and y = f(x^2 + 1) then $\frac{dy}{dx}$ at x = 1 is

(A) 2 sin2 (B) 2 cos 2 (C) 2 sin 4 (D) cos 2

Ans. C

Sol. f'(x) sin (x^2)

y = f(x^2 + 1)

$$\frac{dy}{dx} = f'\ (x^2 + 1)\ 2x$$

$$\frac{dy}{dx} = 2u\ \sin\left((x^2\ +\ 1)^2\right)$$

Put x = 1

$\frac{dy}{dx}$ = 2 sin (4)

Q28. If f(x) = |sin x – |cos x||, then f' $\left(\frac{7\pi}{6}\right)$ =

(A) $\frac{\sqrt{3}+1}{2}$ (B) $\frac{1-\sqrt{3}}{2}$ (C) $\frac{\sqrt{3}-1}{2}$ (D) $\frac{-1-\sqrt{3}}{2}$

Ans. C

Sol. f(x) = | sin (x) – | cos x ||

f(x) = |sin x + cos x|

f(x) = – (sin x + cos x)

f'(x) = – cos x + sin x

Put x = $\frac{7\pi}{6}$

f' $\left(\frac{7\pi}{6}\right) = + \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$

Q29. If f(x) = $\sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$; then f '(0) is equal to

(A) 4 (B) 3 (C) 2 (D) 1

Ans. D

Sol. y = f (x) = $\sqrt{\frac{1\ +\ sin^{-1}(x)}{1-\ tan^{-1}(x)}}$

$$\frac{dy}{dy} = \frac{1}{2}\left(\frac{1\ +\ sin^{-1}(x)}{1\ -\ tan^{-1}(x)}\right)^{\frac{-1}{2}} \left(\frac{\left(1-tan^{-1}(x)\right)\left(\frac{1}{\sqrt{1-x^2}}\right)+(1+sin^{-1}(x)\left(\frac{1}{1+x^2}\right)}{(1-tan^{-1}(x))}^{2}\frac{dy}{dx}\right)_{x=0}$$

= 1

Put a = 0

Q30. If f(x) = 2 + |x| – |x – 1| – |x + 1|, then f' $\left(-\frac{1}{2}\right)+f'\left(\frac{1}{2}\right)+f'\left(\frac{3}{2}\right)+f'\left(\frac{5}{2}\right)$ is equal to

(A) 1 (B) –1 (C) 2 (D) –2

Ans. D

Sol.

$$f(x) \begin{cases} 2-x & x \ge 1 \\ x & 0 < x < 1 \\ -x & -1 \le x \le 0 \\ x+2 & x < -1 \end{cases}$$

$$f'(x) \begin{cases} -1 & x > 1 \\ 1 & 0 < x < 1 \\ -1 & -1 < x < 1 \\ 1 & x < -1 \end{cases}$$

$f'\left(\frac{-1}{2}\right) = -1$

$f'\left(\frac{1}{2}\right) = 1$

$f' = \left(\frac{3}{2}\right) -1$

$f'\left(\frac{5}{2}\right) = -1$

Q31. (a) Let g (x) = ln f (x) where f (x) is a twice differentiable positive function on $(0, \infty)$ such that

f (x + 1) = x f (x). Then for N = 1, 2, 3 $= g''\left(N+\frac{1}{2}\right) - g\, g''\left(\frac{1}{2}\right) =$

(A) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+\ldots\ldots+\frac{1}{(2N-1)^2}\right\}$ (B) $4\left\{1+\frac{1}{9}+\frac{1}{25}+\ldots\ldots+\frac{1}{(2N-1)^2}\right\}$

(C) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+\ldots\ldots+\frac{1}{(2N+1)^2}\right\}$ (D) $4\left\{1+\frac{1}{9}+\frac{1}{25}+\ldots\ldots+\frac{1}{(2N+1)^2}\right\}$

Ans. A

Sol. (a) $g(x+1) = \log(f(x+1)) = \log x + \log f(x)$

$\Rightarrow g(x+1) = \log x + g(x)$

$\Rightarrow g(x+1) - g(x) = \log x$

$\Rightarrow g'(x+1) - g'(x) = \frac{1}{x}$

$\Rightarrow g'(x+1) - g'(x) = \frac{1}{x^2}$

$\Rightarrow g''\left(1+\frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = -4$

$\Rightarrow g''\left(2+\frac{1}{2}\right) - g''\left(1+\frac{1}{2}\right) = -\frac{4}{9}$

$g''\left(N+\frac{1}{2}\right) - g''\left(N-\frac{1}{2}\right) = \frac{-4}{(2N-1)^2}$

By adding

Hence $g''\left(N+\frac{1}{2}\right) - g''\left(\frac{1}{2}\right)$

$= -4\left(1+\frac{1}{9}+\ldots\ldots\ldots\ldots+\frac{1}{(2N-1)^2}\right)$

Q32. Let f : (–1, 1) → R be a differentiable function with f(0) = –1, f '(0) = 1. Let g(x) = |f(2f(x) + 2)|2.]
Then g'(0) :

(A) 4 (B) –4 (C) 0 (D) –2

Ans. B

Sol. $g(x) = [f(2f(x) + 2)]^2$

$g'(x) = 2f(2f(x) + 2)\, f'(2f(x) + 2)\, 2f'(x)$

Put x = 0

$g'(0) = 2f(2f(0) + 2)\, f'(2f(0) + 2)\, 2f'(0)$

$= 2f(2(-1) + 2)\, f'(2(-1) + 2)\, 2f'(0)$

$= 2f(0)\, f'(0)\, 2f'(0)$

$= 4(-1)(1)(1) = -4$

Q33. For $x \in R$, f(x) = |log2–sinx| and g(x) = f(f(x)), then :

(A) g'(0) = – cos (log2) (B) g is differentiable at x = 0 and g'(0) = – sin(log 2)

(C) g is not differentiable at x = 0 (D) g'(0) = cos(log 2)

Ans. D

Sol. g(x) = |log 2 – sin| log 2 – sinx||

gx near by of x = 0

g(x) = log 2 – sin (log 2 – sinx)

g'(x) = – (cos(cos 2–sinx)) (–cosx)

g'(0) = cos(log 2)

Q34. If for $x \in \left(0, \frac{1}{4}\right)$, the derivative of $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x}.\,g$, then g(x) equals :

(A) $\frac{3x}{1-9x^3}$ (B) $\frac{3}{1+9x^3}$ (C) $\frac{9}{1+9x^3}$ (D) $\frac{3x\sqrt{x}}{1-9x^3}$

Ans. C

Sol. $f(x) = \tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right) = 2\tan^{-1}(3x\sqrt{x})$

$\Rightarrow$ f'(x) = $\frac{9\sqrt{x}}{1+9x^3} = \sqrt{x}$. g(x)

$\Rightarrow$ g(x) = $\frac{9}{1\ +\ 9x^3}$

Q35. Let g(x) be the inverse of f'(x) such that f'(x) = $\frac{1}{1+x^5}$, then $\frac{d^2(g(x))}{dx^2}$ is equal to

(A) $\frac{1}{1+(g(x))^5}$ (B) $\frac{g'(x)}{1+(g(x))^5}$ (C) $5(g(x))^4\ (1+(g(x))^5)$ (D) $1+(g(x))^5$

Ans. C

Sol. f' (x) = $\frac{1}{1+x^5}$ $\frac{a^2(g(x)}{cl\ x^2}$

f(x (x)) = x

Differential wrt x

f'(g (x))g' (x) = 1

$\frac{g'(x)}{1+(g(x))^5} = 1$

$g'(x) = 1 + (g(x))^5$

Diff. wrt x

$g''(x) = 5(g(x))^4\ (g'(x))$

$= 5\ (g(x)^4)\left(1+(g(x))^5\right)$

Q36. Let f (x) = x + sin x. Suppose g denotes the inverse function of f. The value of g' $\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)$ has the value equal to

(A) $\sqrt{2}-1$ (B) $\frac{\sqrt{2}-1}{\sqrt{2}}$ (C) $2-\sqrt{2}$ (D) $\sqrt{2}+1$

Ans. C

Sol. f (x) = y = x + sin x

$\frac{dy}{dx} = 1 + \cos x$

g'y = $\frac{dy}{dx} = \frac{1}{1+cos\ x}$

where y = $\frac{\pi}{4}+\frac{1}{\sqrt{2}}$= x + sin x $\Rightarrow$ x $=\frac{\pi}{4}$

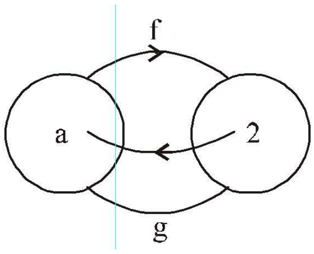
$\therefore$ g' $\left(\frac{\pi}{2}+\frac{1}{\sqrt{2}}\right) = \frac{1}{1\ +\ (1/\sqrt{2})} = \frac{\sqrt{2}}{\sqrt{2}+1} = \sqrt{2}(\sqrt{2}-1) = 2-\sqrt{2}$

Q37. Let g is the inverse function of f and f' (x) = $\frac{x^{10}}{(1+x^2)}$.

(A) $\frac{5}{2^{10}}$ (B) $\frac{1+a^2}{2^{10}}$ (C) $\frac{a^{10}}{1+a^2}$ (D) $\frac{1+a^{10}}{a^2}$

Ans. B

Sol.



f [g(x)] = x $\Rightarrow$ f' [g (x)] . [g'(x)] = 1 $\Rightarrow$ f ' [g(2)] g ' (2) = 12 $\Rightarrow$ f ' (a) g ' (2) [putting x = 2]

given, f' (a) = $\frac{a^{10}}{1\ +\ a^2}$ $\Rightarrow$ g' (2) = $\frac{1\ +\ a^2}{a^{10}}$]

Alternative : g [f (x)] = x

g' [f (x)] . f ' (x) = 1

now g (2) = a $\Rightarrow$ f (a) = 2

$\therefore$ g and f are inverse of each other

now f (x) = 2 $\Rightarrow$ g (2) = x = a

$\therefore$ g ' (2) . f ' (a) = 1

g' (2) = $\frac{1}{f'(a)} = \frac{1+a^2}{a^{10}}$

Q38. If y =$\frac{1}{x^2-a^2}$, then $\frac{d^y}{dx^2}$ equals-

(A) $\frac{3x^2+a^2}{(x^2-a^2)^3}$ (B) $\frac{3x^2+a^2}{(x^2-a^2)^4}$ (C) $\frac{2(3x^2+a^2)}{(x^2-a^2)^3}$ (D) $\frac{2(3x^2+a^2)}{(x^2-a^2)^4}$

Ans. C

Sol. $$y = \frac{(x\ +\ a)\ -\ (x\ -\ a)}{2a\ (x\ +\ a)(x\ -\ a)}$$

$$y = \frac{1}{2a}\left(\frac{1}{x\ -\ a} - \frac{1}{x\ +\ a}\right)$$

$$\frac{dy}{dx} = \frac{1}{2a}\left(\frac{-1}{(x-a)} + \frac{1}{(x\ +\ a)^2}\right)$$

$$\frac{d^2y}{dx^2} = \frac{1}{2a}\left(\frac{2}{(x-a)^3} + \frac{2}{(x\ +\ a)^3}\right)$$

$$\frac{1}{a}\left(\frac{(x\ +\ a)^3\ -\ (x\ -\ a)^3}{(x\ +\ a)^3\ (x\ -\ a)^3}\right)$$

$$= \frac{2(3x^2\ +\ a^2)^{\ 3}}{(x^2\ -a^2)}$$

Q39. If y = ae$^{\alpha x}$ – be –$^{\alpha x}$, then the value of $\frac{d^2y}{dx^2}$ equals –

(A) αy (B) $-\alpha y$ (C) $\alpha^2 y$ (D) $-\alpha^2 y$

Ans. C

Sol. y = ae$^{\alpha x}$ – be$^{\alpha x}$

$\frac{dy}{dx}$ = be $a\alpha\, e^{\alpha x} + \ a\alpha e^{-\alpha x}$

$\frac{d^2}{dx^2}$ aα^2 $e^{\alpha^x} - b^{\alpha^2}\ e^{-\alpha^x}$

= α^2y

Q40. If y$^{1/m}$ = x + $\sqrt{1+x^2}$, then (1 + x^2) $\frac{d^2y}{dx^2}$+ x$\frac{dy}{dx}$ equals –

(A) my^2 (B) m^{2}y (C) m^{2}y^2 (D) None of these

Ans. B

Sol. y$^{1/m}$ = x + $\sqrt{1+a}^2$

$\frac{dy}{dx}\left(\frac{1}{m}y^{\frac{1}{m}-1}\right) = 1 + \frac{x}{\sqrt{1+\ x^2}}$

$\frac{dy}{dx}\left(\frac{y^{\frac{1}{m}-1}}{m}\right) = \frac{\sqrt{1\ +\ x^2}\ +\ x}{\sqrt{1+\ a^2}}$

$\frac{dy}{dx}\left(\frac{y^{\frac{1}{m}-1}}{m}\right) = \frac{y^{\frac{1}{m}}}{\sqrt{1+x^2}}$

$\sqrt{1+x^2} = \frac{dy}{dx}$ = my

Diff. wrt v

$\frac{x}{\sqrt{1+x^2}}\ \frac{dy}{dx} + \sqrt{1+x^2}\ \frac{dy^2}{dx^2}$ = m $\frac{dy}{dx}$

(1 + x^2) $\frac{d^2y}{dx^2} + \frac{xdy}{dx}\ m\sqrt{1+\ x^2}\ \frac{dy}{dx}$

= m (my)

= m^{2}y

Q41. If y = a cos (log x) + b sin (log x) then (x^{2}y$_2$ + xy$_1$) equals to –

(A) y (B) ay (C) by (D) –y

Ans. D

Sol. y = a cos (log x) + b sin (log x)

Diff. wrt x

$\frac{dy}{dx} = \frac{-a}{x}$ sin (log x) + $\frac{b}{x}$ cos (log x)

x $\frac{dy}{dx}$ = a sin (log x) + b cos (log x)

Diff. wrt x

x $\frac{d^2y}{dx^2} = \frac{dy}{dx} = \frac{-\, a\, cos\, (log\, x)}{x} + \frac{b\, sin\, (log\, x)}{x}$

$x^2y_{2} + y_1$ = – (a cos (log x) + b sin (log x))

= – y

Q42. If y = sin 2x cos 6x, then the value of $\frac{d^2y}{dx^2}$ is–

(A) 8 sin 4x + 32 sin 8x (B) 32 sin 8x – 8 sin 4x (C) 8 sin 4x – 32 sin8x

(D) None of these

Ans. C

Sol. y = $\frac{1}{2}$(sin 8x – sin 4x)

Diff. wrt x

$\frac{dy}{dx} = \frac{1}{2}$ (8 cos 8x – 4 cos 4x)

Diff. wrt x

$\frac{d^2y}{dx^2} = \frac{1}{2}$ (– 64 sin 8x + 16 sin 4x)

= 8 sin 4x – 32 sin 8x

Q43. If y = $\sqrt{2x-x^2}$, then $yy_2 + y_1^2$ is equal to

(A) 0 (B) –1 (C) 1 (D) Does not exist

Ans. B

Sol. $y = \sqrt{2x-x^2}$

$y^2 = 2x - x^2$

$2yy_1 = 2 - 2x$

$2yy_1 = 2 - 2x$

$yy_1 = 1 - x$

Diff. wrt x

$yy_2 + (y_1)^2 = -1$

Q44. If f(x) = x^n, then the value of $f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + \ldots + \frac{(-1)^n f^n(1)}{n!}$ is-

(A) 1 (B) 2^n (C) 2^{n-1} (D) 0

Ans. D

Sol. $|(x) = x^n$

$\therefore |'(x) = nx^{n-1}, |''(x) = n(n-1)x^{n-2}$

....

$|''^{\,\ldots\ldots n \text{ times}}(x) = n!$

Now $|(1) - \frac{|'(1)|}{1!} + \frac{|'(1)|}{2!} - \ldots\ldots\ldots \frac{|''''^{\ldots\ldots n\ times}(1)}{n!}$

$= 1 - \frac{n}{1!} + \frac{n(n-1)}{2!} - \ldots\ldots\ldots = \frac{n!}{n!} = (1-1)^n = 0$

Q45. Let 'f' be a differentiable real valued function satisfying f(x + 2y) = f(x) + f(2y) + 6xy (x + 2y) ∀ x, y ∈ R. Then
f ''(0), f ''(1), f ''(2) .... are in

(A) AP (B) GP (C) HP (D) None of these

Ans. A

Sol. f (x + 2y) = f(x) + f(2y) + 6xy (x + 2y)

$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ f'(x) = k

$\frac{f(x) + f(h) + 3xh(x+h) - f(x)}{h}$

$= \lim_{h\to 0}\left(\frac{f(h)}{h} + 3x(x+h)\right)$

$f'(x) = f'(0) + 3x^2$

$f''(x) = 6x$

$f''(0) = 0$

$f''(0) = 6$

$f''(2) = 12$ A.P.

Q46. $\frac{d^2x}{dy^2}$ equals

(A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (B) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$ (C) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (D) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

Ans. D

Sol. $\frac{d}{dy} = \left(\frac{dx}{dy}\right) = \frac{d}{dy}\left(\frac{d}{dy/dy}\right)$

$$= -\left(\frac{dy}{dx}\right)^{-2}\frac{d^2y}{dx^2}.\frac{dx}{dy} = -\frac{d^2y}{dx^2}\left(\frac{dy}{dx}\right)^{-3}$$

Q47. Let $f(x) = \frac{e^{\tan x} - e^x + ln(\sec x + \tan x) - x}{\tan x - x}$ be a continuous function at x = 0. The value of f (0) equal

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 2

Ans. C

Sol. For continuity $k = f(x) = \lim_{x\to 0} f(x) = \lim_{x\to 0}\frac{e^{\tan x} - e^x}{\tan x - x} + \lim_{x\to 0}\frac{ln(\sec x + \tan x) - x}{\left(\frac{\tan x - x}{x^3}\right).x^3}$

$$= \lim_{x\to 0}\frac{e^x(e^{\tan x - x} - 1)}{\tan x - x} + 3\lim_{x\to 0}\frac{ln(\sec x + \tan x) - x}{x^3}$$

$$= 1 + 3\lim_{x\to 0}\frac{\sec x - 1}{3x^2} = 1 + \frac{1}{2} = \frac{3}{2}$$

Q48. If the normal of y = f(x) at (0, 0) is given by y – x = 0, then

$\lim_{x\to 0}\frac{x^2}{f(x^2) - 20f(9x^2) + 2f(99x^2)}$ is equal to

(A) $\frac{1}{19}$ (B) $\frac{-1}{19}$ (C) $\frac{1}{2}$ (D) non-existent

Ans. B

Sol. The form being $\frac{0}{0}$, so the required limit by L' Hospital rule is [12th, 25-07-2010 P-2]

$$\lim_{x\to 0}\frac{2x}{2x\ f'(x)^2 - 360f(9x^2) + 396\ f(99x^2)} = \lim_{x\to 0}\frac{1}{2x\ f'(x^2) - 180f(9x^2) + 198\ f(99x^2)}$$

For f (x) = 0, the slope of tangent is f '(x) and slope of normal at x = 0 is

$\frac{-1}{f'(0)}$ = 1 (Given)

$\Rightarrow$ f ' (0) = – 1

$\therefore$ Limit = $\frac{1}{f'(0) - 180f'(0) \ + \ 198f'(0)}$ = $\frac{-1}{(1 - \ 180 \ + \ 198)}$ = $\frac{-1}{19}$ Ans.]

Q49. $\lim_{x\to\infty}\left(\frac{x^{100}}{e^x} + \left(\cos\frac{2}{x}\right)^{x^2}\right)$ equals

(A) e^{-1} (B) e^{-4} (C) $(1+ e^{-2})$ (D) e^{-2}

Ans. D

Sol. $\lim_{x\to\infty}\frac{x^{100}}{e^x}$= 0 (using L' Hospital's Rule)

$\lim_{x\to\infty}\left(cos\frac{2}{x}\right)^{x^2}$ (1^{∞})

$e^{\lim_{x\to\infty} x^2\left(cos\frac{2}{x}-1\right)} = e\ e^{\lim_{x\to 0}\left(\frac{1-cos\, t}{t^2}.4\right)}$; put $\frac{2}{x}$= t $\Rightarrow$ x = $\frac{2}{t}$

= e^{-2}

Q50. Let f (x) be diffrentiable at x = h then $\lim_{x\to h}\frac{(x+h)f(x) - 2hf(h)}{x-h}$ is equal to

(A) f(h) + 2hf '(h) (B) 2 f(h) + hf '(h) (C) hf(h) + 2f '(h) (D) hf(h) – 2f '(h)

Ans. A

Sol. Use L' Hospital's rule

Q51. For x > 0, $\lim_{x\to 0}\left((\sin x)^{1/x} + (1/x)^{\sin x}\right)$ is

(A) 0 (B) –1 (C) 1 (D) 2

Ans. C

Sol. $\lim_{x\to 0}\left((sinx)^{1/x} + (\frac{1}{x})^{sin\, x}\right) = \lim_{x\to 0}(sin\ x)^{1/x} + \lim_{x\to 0} x^{sin\ x}$

= L + M

L $= \lim_{x\to 0}(sin\ x)^{1/x} = 0$ (0^{∞} form)

M = $\lim_{x\to 0} x^{-sin\ x} = \lim_{x\to 0} e^{-sin\ x\ ln\ x}$

= $e^{\lim_{x\to 0} - \lim_{x\to 0}}$ e^{p}

P = $\lim_{x\to 0}\ - \frac{cos\ x}{cosec\ x}$

hospital Rule

= $\lim_{x\to 0}\ \frac{-\frac{1}{x}}{-\ cosec\ x\ cot\ x}$

$$= \lim_{x\to 0} \frac{sin\ x\ tan\ x}{x} = 0$$

$M = e^0 = 1$

Ans. = L + M = 0 + 1 = 1

Exercise J2

Q1. Let f , g and h are differentiable functions. If $f(0) = 1$; $g(0) = 2$; $h(0) = 3$ and the derivatives of their pair wise products at x = 0 are (f g)'(0) = 6 ; (g h)'(0) = 4 and (h f)'(0) = 5 then compute the value of (fgh)'(0).

Ans. 16

Sol.

$$(fgh)'(x) = \frac{h(x)(fg)'(x) + f(x)(gh)'(x) + g(x)\Big((hf)'(x)\Big)}{2}$$

Put 4 = 0

$$(fgh)(0) = \frac{(3 \times 6) \; + \; (1 \times 4) \; + \; (2 \times 5)}{2} = 16$$

Q2. (a) If $y = (\cos x)^{\ln x} + (\ln x)^x$ find $\frac{dy}{dx}$ (b) If $y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$. Find $\frac{dy}{dx}$.

Ans. (a) $Dy = (\cos x)^{\ln x}\left[\frac{ln(cosx)}{x} - \tan x lnx\right] + (lnx)^x\left[\frac{1}{lnx} + ln(lnx)\right]$;

(b) $\frac{dy}{dx} = e^{x^{e^x}} . x^{e^x}\left[\frac{e^x}{x} + e^x lnx\right] + e^{x^{x^e}} x^{e-1} e^{x^e}[1+e\ln x] + x^{e^{e^x}} e^{e^x}\left[\frac{1}{x} + e^x lnx\right]$

Sol. (a) $y = (\cos x)^{\ln x} + (\ln x)^x$

$y = a + b$

$a = (\cos x)^{\ln x}$

$\frac{da}{dx} = (\cos x)^{\ln x}\left((ln\ x)\ (-\ ln\ x) + \frac{ln\ cos\ x}{x}\right)$

$b = (\ln x)^x$

$\frac{dy}{dx} = \frac{da}{dx} + \frac{db}{dx}$

(b) y = f(x) + g(x) + h(x)

$f(x) = e^{x^{e^x}}$

$\ln f(x) = x^{e^x}$

$\ln\left(ln\ (f\ (x)\right) = e^x \ln x$

Diff.

$\frac{f'(x)}{f(x)\ ln\ f(x)} = \frac{e^x}{x} + (\ln x)e^x$

$f'(x) = f(x) \ln f(x)\left(\frac{e^x}{x} \; + \; (ln\ x)e^x\right)$

$= e^{x^{e^x}}\ x^{e^x}\left(\frac{e^x}{x} \; + \; ln\ x\right)$

g (x) = $e^{x^{e^x}}$

ln g (x) = $x^{x^{x^e}}$

ln (ln g (x)) = x^e ln x

Diff. wrt x

$\frac{g'(x)}{g(x)\ ln\ g\ (x)} = \frac{x^e}{x}$ + ex^{e-1} ln x

g'(x) = g(x) ln g(x) $\left(x^{e-1}(1\ +\ e\ ln\ x)\right)$

g'(x) = $e^{x^{x^e}}$ $e^{x^{x^e}}$ (x^{e-1}) (1 + e ln x)

h (a) = $e^{e^{e^x}}$

ln h (x) = e^{e^x} ln x

$\frac{h'(x)}{h(x)} = \frac{e^{e^x}}{x}$ + e^{e^x} e^x ln x

h' (x) = h(x) $\left(\frac{e^{e^x}}{x} + e^{e^x} e^x\ ln\ x\right)$

$e^{e^{e^x}} e^{e^x} \left(\frac{1}{x} + e^x\ ln\ x\right)$

y = f(x) + g(x) + h(x)

$\frac{dy}{dx}$= f'(x) + g'(x) + h'(x)

Q3. If y = $\frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + ln\sqrt{x+\sqrt{x^2+1}}$ prove that 2y = xy' + ln y'. where ' denotes the derivative.

Ans. -

Sol. y = $\frac{x^2}{2} + \frac{1}{2}$ x$\sqrt{x^2\ +1}$ + ln $\sqrt{x\ +\ \sqrt{x^2+1}}$

2y x^2 + x$\sqrt{x^2\ +1}$ + ln $\left(x\ +\ \sqrt{x^2+1}\right)$ ....(i)

2y' = 2x + x $\sqrt{x^2\ +1}$ + $\frac{2x^2}{2\sqrt{x^2\ +\ 1}}$ + $\frac{1+\frac{2x}{2\sqrt{x^2+1}}}{x^2+\sqrt{x^2+1}}$

2y' = 2x + x $\sqrt{x^2\ +1}$ +$\frac{x^2}{x^2+1}$ + $\frac{1}{\sqrt{x^2+1}}$

= 2x + $\sqrt{x^2\ +1}$ + $\sqrt{x^2\ +1}$

y' = (x + $\sqrt{x^2\ +1}$)

Q4. If $y = \ln\left(x^{e^x.a^y}\right)^{y^x}$ find $\frac{dy}{dx}$.

Ans. $\frac{y}{x}.\frac{x\ell nx + x\ell nx.\ell ny + 1}{\ell nx(1–x–y\ell na)}$

Sol. $y = y^x \ln \ln (e^{e^x a^y})$

$y = y^x e^x a^y \ln x$

Take logarithm ln y = x ln y + x + y ln a + ln (ln x)

Diff. wrt x

$$\frac{1}{y}\frac{dy}{dx} = \ln + \frac{x}{y}\frac{dy}{dx} + 1 \text{ (ln a) } \frac{dy}{dx} + \frac{1}{x\, ln\,(x)}$$

$$\frac{dy}{dx}\left(\frac{1}{y}\frac{x}{y} - ln\,x\right) = 1 + \frac{1}{x\,ln\,x} + \ln y$$

$$\left(\frac{1 - x - \,yln\,x}{y}\right)\frac{dy}{dx} = \frac{x\,ln\,x \;+\; 1\,x\,lnx\,ln\,y}{x\,ln\,x}$$

$$\frac{dy}{dx} = \left(\frac{x\,ln\,x \;+\; 1\,x\,lnx\,ln\,y}{x\,ln\,x}\right)$$

Q5. If $y = 1 + \frac{a_1}{x - a_1} + \frac{a_2 .\, x}{(x - a_1)(x - a_2)} + \frac{a_3 x}{(x - a_1)(a - x)(x - a_3)} + \ldots$ upto (n + 1) terms then prove that

$$\frac{dy}{dx} = \frac{y}{x}\left[\frac{a_1}{a_1 - x} + \frac{a_2}{a_2 - x} + \frac{a_3}{a_3 - x} + \ldots + \frac{a_n}{a_n - x}\right]$$

Ans. -

Sol. $y = 1 + \frac{a_1}{x - a_1} + \frac{a_2 x}{(x - a_1)(x - a_2)} + \frac{a_3 x}{(x - a_1)(x - a_2)(x - a_3)} + \ldots$ (n + 1) terms

$= \frac{x}{(x - a_1)} + \frac{x}{(x - a_1)} + \frac{a_3 x}{(x - a_1)(x - a_2)(x - a_3)} + \ldots\ldots + $ (n + 1) terms.

$= \frac{x^2}{(x - a_1)(x - a_2)} + \frac{a_3 x}{(x - a_1)(x - a_2)(x - a_3)} + \ldots$ (n + 1) terms, finally we get

$$y = \frac{x^n}{(x - a_1)(x - a_2)(x - a_3)\ldots\ldots(x - a_n)}$$

Taking $\log_e$ of both sides, we have

$\ln y = n\, \ln x - \ln (x - a_1) - \ln (x - a_2) - \ln (x - a_3) - \ldots\ldots\ldots\ldots - \ln (x - a_n)$

Differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \left[\frac{n}{x} - \frac{1}{x - a_1} - \frac{x}{x - a_2} - \frac{x}{x - a_3} - \ldots - \frac{x}{x - a_n}\right]$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \left[n - \frac{1}{x - a_1} - \frac{x}{x - a_2} - \frac{x}{x - a_3} - \ldots - \frac{x}{x - a_n}\right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}\left[n - \frac{x}{x - a_1} - \frac{x}{x - a_2} - \frac{x}{x - a_3} - \ldots\ldots - \frac{x}{x - a_n}\right]$$

$$= \frac{y}{x}\left[(1+1+1+1+\ldots..n\ times) - \frac{x}{x-a_1} - \frac{x}{x-a_2} - \frac{x}{x-a_3} - \ldots\ldots\ldots - \frac{x}{x-a_n}\right]$$

$$= \frac{x}{y}\left[\left(1 - \frac{x}{x-a_1} + \left(1 - \frac{x}{x-a_2}\right) + \left(1 - \frac{x}{x-a_3}\right) + \ldots.. + \left(1 - \frac{x}{x-a_n}\right)\right]$$

$$\therefore \frac{dy}{dx} = \frac{y}{x}\left[\left(\frac{-a_1}{x-a_1}\right) + \left(\frac{-a_2}{x-a_2}\right) + \left(\frac{-a_3}{x-a_3}\right) + \ldots\ldots\ldots\ldots + \left(\frac{-a_n}{x-a_n}\right)\right]$$

$$= \frac{y}{x}\left[\frac{a_1}{a_1+x} + \frac{a_2}{a_2+x} + \frac{a_3}{a_3+x} + \ldots\ldots. + \frac{a_n}{a_n+x}\right]$$

Q6. If $x = \text{cosec}\ \theta - \sin\theta$; $y = \text{cosec}^n\ \theta - \sin^n\theta$, then show that $(x^2+4)\left(\frac{dy}{dx}\right)^2 - n^2(y^2 + 4) = 0$.

Ans. -

Sol. $x = \text{cosec}\ \theta - \sin\theta$

$$\frac{dx}{d\theta} = -\text{cosec}\ \theta \cot\theta - \cos\theta$$

$$\frac{dx}{d\theta} = -\cot\theta\ (\text{cosec}\ \theta + \sin\theta)$$

$y -\ \text{cosec}^x\ (\theta) - (\sin\theta)^x$

$$\frac{dy}{d\theta} = -n\ \text{cosec}^n\ (\theta) \cot\theta - n(\sin\theta)^{n-1}\cos\theta$$

$$= -n\cos\theta\ \left((cosec\ \theta)^n + (sin\ \theta)^n\right)$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{n\left((cosec\ \theta)^n + (sin\ \theta)^n\right)}{cosec\ \theta + sin\ \theta}$$

$$\therefore a + b\sqrt{(a+b)^n + 4ab}$$

$$\frac{dy}{dx} = \frac{n\sqrt{\left((cosec\ \theta)^n - (sin\ \theta)^n\right)^2 + 4}}{\sqrt{(cosec\ \theta -\ \theta)^2 + 4}}$$

$$\frac{dy}{dx} = \frac{n^2\sqrt{y^2 + 4}}{x^2 + 4}$$

sauaring

$$\left(\frac{dy}{dx}\right)^2 \frac{n^2\sqrt{(y^2 + 4)}}{x^2 + 4}$$

Q7. If a curve is represented parametrically by the equations
$x = \sin\left(t + \frac{7\pi}{12}\right) + \sin\left(t + \frac{\pi}{12}\right) + \sin\left(t + \frac{3\pi}{12}\right)$, $y = \cos\left(t + \frac{7\pi}{12}\right) + \cos\left(t - \frac{\pi}{12}\right) + \cos\left(t + \frac{3\pi}{12}\right)$ then find the value of at $t = \frac{d}{dt}\left(\frac{x}{y} - \frac{y}{x}\right)$ at $t = \frac{\pi}{8}$.

Ans. 8

Sol. $x = \sin\left(t + \frac{7\pi}{12}\right) + \sin\left(t + \frac{\pi}{12}\right) + \sin\left(t + \frac{\pi}{4}\right)$

$x = 2\sin\left(t + \frac{\pi}{12}\right)\cos\frac{\pi}{12} + \sin\left(t + \frac{\pi}{4}\right)$

$= \sin\left(t + \frac{\pi}{4}\right) + \sin\left(t + \frac{\pi}{4}\right)$

$x = 2\sin\sin\left(t + \frac{\pi}{4}\right)$

lly $y = \cos\sin\left(t + \frac{\pi}{4}\right)$

$\frac{1}{y}\frac{dy}{dx} = \ln y + \frac{x}{y}\frac{dy}{dx} + 1 + (\ln a)\frac{dy}{dx} + \frac{1}{x\,ln(x)}$

$\frac{dy}{dx}\left(\frac{1}{y} - \frac{x}{y} - (ln\,a)\right)\frac{dy}{dx} - \left(\frac{x\,ln\,x + 1 + x\,ln\,x\,ln\,y}{x\,ln\,x}\right)$

$\left(\frac{1 - x - y\,ln\,a}{y}\right)\frac{dy}{dx} - \left(\frac{x\,ln\,x + 1 + x\,ln\,x\,ln\,y}{x\,ln\,x}\right)$

$\frac{dy}{dx} - \left(\frac{x\,ln\,x + 1 + x\,ln\,x\,ln\,y}{x\,ln\,x}\right)$

$E = \frac{x}{y} - \frac{y}{x} = \tan\left(t + \frac{\pi}{4}\right) - \cot\left(t + \frac{\pi}{4}\right)$

$= -2\cot\left(2t + \frac{\pi}{4}\right)$

$E = 2\tan 2t$

$\frac{dE}{dt} = 4\sec^2(2t)$

Put $t = \frac{\pi}{8}$

$\frac{dE}{dt} = 4\sec^2\left(\frac{\pi}{4}\right) = 4(1) = 8$

Q8. If $x = \frac{1 + lnt}{t^2}$ and $y = \frac{3 + 2lnt}{t}$. Show that $y\frac{dy}{dx} = 2x\left(\frac{dy}{dx}\right)^2 + 1$.

Ans. -

Sol. $x = \frac{1 + ln\,t}{t^2}$ $\qquad$ $y = \frac{3 + 2\,ln\,t}{t}$

$\frac{dx}{dt} = \frac{t^2\left(\frac{1}{t} - (1 + ln\,t)(2t)\right)}{t^4}$

$= \frac{t(1 - 2 - 2\,ln\,t)}{t^4}$

$= -\frac{(1 + 2\,ln\,t)}{t^3}$

$\frac{dy}{dx} = \frac{\frac{fu}{fy}}{\frac{dx}{dt}} = t$

$\therefore x = \frac{1 + \ ln\, t}{t^2}$

$t^2 x = 1 + \ln t$

eliminated ln t

$ty = 2t^2 x + 1$

put y $\frac{dy}{dx} = 2x \left(\frac{dy}{dx}\right)^2 + 1$

$y = \frac{3 + 2\, ln\, t}{t}$

$\frac{dy}{dx} = \frac{t\left(\frac{2}{t}\right) - (3 + 2\, ln\, t)}{t^2}$

$\frac{2 - 3 - 2\, ln\, t}{t^2}$

$= -\frac{(1 + 2\, ln\, t)}{t^3}$

$y = \frac{3 + 2\, ln\, t}{t}$

$ty = 3 + 2 \ln t$

eliminated ln t

$ty = 2t^2 x + 1$

put y $\frac{dy}{dx} = 2x \left(\frac{dy}{dx}\right)^2 + 1$

Q9. Differentiate $\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ w. r. t. $\sqrt{1-x^4}$.

Ans. $\frac{1 + \sqrt{1-x^4}}{x^6}$

Sol. $y = \frac{\sqrt{1\ +\ x^2} + \sqrt{(1-x)}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ $t = \sqrt{1-x^4}$

$\frac{dy}{dt} = \frac{\frac{dy}{dx}}{\frac{dt}{dx}}$

Q10. Let g(x) be a polynomial, of degree one & f(x) be defined by $f(x) = \left[\begin{array}{ll} g(x), & x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x} & x > 0 \end{array}\right.$

.

Find the continuous function f(x) satisfying $f'(1) = f(-1)$

Ans.
$$f(x) = \begin{cases} -\frac{2}{3}\left[\frac{1}{6} + ln\frac{3}{2}\right]x & if \quad x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x} & if \quad x > 0 \end{cases}$$

Sol. for x > 0

$$f(x) = \left(\frac{1 + x}{2 + x}\right)^{1/2}$$

$$\ln f(x) = \frac{1}{x}\left(ln\,(1+x) - ln(2+x)\right)$$

x ln f(x) = ln (1 + x) – ln (2 + x)

Diff. wrt x

$$\ln f(x)\ \frac{x}{f(x)} = f'(x) = \frac{1}{1+x} - \frac{1}{2 + x}$$

Put x = 1 & $f(1) = \frac{2}{3}$

$$\ln\left(\frac{2}{3}\right) + \frac{f'(1)}{\frac{2}{3}} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$\frac{2}{3}f'(1) = \frac{1}{6} + ln\left(\frac{3}{2}\right)$$

$$f'(1) = \frac{2}{3}\left(\frac{1}{6} + ln\left(\frac{3}{2}\right)\right)$$

let g (x) = ax + b

f(x) is centinuous at x = 0

$$g(0) = \lim_{x\to 0} f(x)$$

f'(1) = f (– 1) = g (– 1)

$$-a = \frac{2}{3}\left(\frac{1}{6} + ln\left(\frac{3}{2}\right)\right)$$

$$a = \frac{-2}{3}\left(\frac{1}{6} + ln\left(\frac{3}{2}\right)\right)$$

⇒ g (x) = ax + b

$$a = -\frac{2}{3}\left(\frac{1}{6} + ln\left(\frac{3}{2}\right)\right)$$

$$\Rightarrow f(x) \begin{cases} -\frac{2}{3}\left(\frac{1}{6}+\ln\left(\frac{3}{2}\right)\right)x & x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x} & x > 0 \end{cases}$$

Q11. If $\sqrt{1-x^6}+\sqrt{1-y^6}=a^3.(x^3-y^3)$ prove that $\frac{dy}{dx}=\frac{x^2}{y^2}\sqrt{\frac{1-y^2}{1-x^6}}$.

Ans. -

Sol. $\sqrt{1-x^6}+\sqrt{1-y^6}=a^3\ (x^3-y^3)$

$x^3=\sin\theta \qquad y^3=\sin\phi$

$\cos\theta+\cos\phi=a^3\ (\sin\theta-\sin\phi)$

$2\cos\left(\frac{\theta+\phi}{2}\right)=a^3\left(2\,sin\left(\frac{\pi}{\phi}\right)cos\left(\frac{\theta+\phi}{2}\right)\right)$

$\cot\left(\frac{\theta-\phi}{2}\right)=a^3$

$\frac{\theta-\phi}{2}=\cot^{-1}(a^3)$

$\theta-\phi=2\cot^{-1}(a^3)$

$\sin^{-1}(x^3)-\sin^{-1}(y^3)=2\cot^{-1}=(a^3)$

Diff. wrt x

$\frac{3x^2}{\sqrt{1-x^6}}-\frac{3y^2}{\sqrt{1-y^6}}\frac{dy}{dx}=0$

$\frac{dy}{dx}=\frac{x^2\sqrt{1-y^6}}{y^2\sqrt{1-x^6}}$

Q12. If $y=x+\cfrac{1}{x+\cfrac{1}{x+\cfrac{1}{x+\ldots\ldots\ldots}}}$, prove that $\frac{dy}{dx}=\cfrac{1}{2\cfrac{x}{x+\cfrac{1}{x+\cfrac{1}{x+\ldots\ldots\ldots}}}}$.

Ans. -

Sol. $y=x+\frac{1}{y}$ to prove $\frac{dy}{dx}=\frac{1}{2\frac{x}{y}}$

$y^2-1=xy$

$2y\frac{dy}{dx}=x\frac{dy}{dx}+y$

$(2y-x)\frac{dy}{dx}=y$

$\frac{dy}{dx}=\frac{y}{2y-x}=\frac{1}{2-\frac{x}{y}}$

Q13. Let $f(x)=x+\frac{1}{2x+}\frac{1}{2x+}\frac{1}{2x+}\ldots\ldots\ldots\infty$. Compute the value of $f(100)\cdot f'(100)$.

Ans. 100

Sol. $y = x + \frac{1}{2x + \frac{1}{2x+.........}}$

$x + y = 2x + \frac{1}{x + y}$

$(x + y)^2 = 2x\,(x + y) + 1$

$(x + y)\,(x + y - 2x) = 1$

$y^2 - x^2 = 1$

$2y\,\frac{dy}{dx} - 2x = 0$

$y\,\frac{dy}{dx} = x$

put x = 100

f(100) f'(100) = 100

Q14. Find the derivative with respect to x of the function: $(\log_{\cos x} \sin x)\,(\log_{\sin x} \cos x)^{-1} + \arcsin \frac{2x}{1+x^2}$ at $x = \frac{\pi}{4}$.

Ans. $\frac{32}{16+\pi^2} - \frac{8}{ln2}$

Sol. $y = \frac{log_{cos\,x}\,sin\,x}{log_{sin\,x} cos\,x} \sin^{-1}\left(\frac{2c}{1\,+\,x^2}\right)$

$y = \left(\frac{log\,sin\,x}{log\,cos\,x}\right)^2 + 2\tan^{-1}(x)$

$\frac{dy}{dx}\;\frac{2\,log\,(sin\,x)}{(log\,cas\,x)}\left(\frac{cot\,x\,log\,cos\,x\;+\;tan\,x\,log\,(sin\,x)}{(log\,cos\,x)}\right)$

$+\frac{2}{1+x}$

put $x = \frac{\pi}{4}$

$\frac{dy}{dx} = -\frac{8}{ln\,2} + \frac{8}{6+\pi^2}$

Q15. Suppose $f(x) = \tan\left(\sin^{-1}(2x)\right)$

(a) Find the domain and range of f.

(b) Express f (x) as an algebaric function of x.

(c) Find $f\,'(1/4)$.

Ans. (a) $\left(-\frac{1}{2}, \frac{1}{2}\right), (-\infty, \infty)$ (b) $f(x) = \frac{2x}{\sqrt{1-4x^2}}$; (c) $\frac{16\sqrt{3}}{9}$

Sol. (a) $f(x) = \tan(\sin^{-1}(2x))$

(a) – 1 < 2x < 1

$\Rightarrow$ x $\in \left(\frac{-1}{2}, \frac{1}{2}\right)$

Range

y $\in$ tan $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

y $\in$ (– ∞, ∞)

(b) $\sin^{-1}$ (2n) = 0

sin θ = 2x

f(x) = tan θ

$= \frac{sin\ \theta}{cos\ \theta} = \frac{sin\ \theta}{\sqrt{1 - \ sin^2\ \theta}}$

f(x) = $\frac{2x}{\sqrt{1-4x^2}}$

(c) f(x) = tan ($\sin^{-1}$ (2x))

f'(x) = $\sec^2$ ($\sin^{-1}$ (2x)) $\left(\frac{1}{\sqrt{1-4x^2}}\right)^2$

Put x = $\frac{1}{4}$

f' $\left(\frac{1}{4}\right) = \frac{16\sqrt{3}}{9}$

Q16. If y =$\tan^{-1} \frac{1}{x^2+x+1}$+$\tan^{-1}$+$\frac{1}{x^2+3x+3}$+$\tan^{-1}$+$\frac{1}{x^2+5x+7}$+$\tan^{-1}\frac{1}{x^2+7x+13}$+......
to n terms.
Find dy/dx , expressing your answer in 2 terms.

Ans. $\frac{1}{1+(x+n)^2} - \frac{1}{1+x^2}$

Sol. $y = \left(tan^{-1}\ + (x+1) - tan^{-1}(x)\right) + \ \left(tan^{-1}\ + (x+2) - tan^{-1}(x+1)\right) +$ ....

.... $\left(tan^{-1}\ + (x+n) - tan^{-1}(x+n-y)\right)$

y = $\tan^{-1}$ (x + n) – $\tan^{-1}$ (x)

$\frac{dy}{dx} = \frac{1}{1+(x+n)^2} + \frac{1}{1\ +\ x^2}$

Q17. If y = $\tan^{-1}\frac{u}{\sqrt{1-u^2}}$ & x =$\sec^{-1}\frac{1}{2u^2-1}$, u $\in \left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$ prove that 2 $\frac{dy}{dx}$+
1 = 0.

Ans. -

Sol. u = cos θ

$$y = \tan^{-1}\left(\frac{cos\ \theta}{sin\ \theta}\right)$$

$$= \frac{\pi}{2} - \ \cos t^{-1} \cot\ \ \theta$$

$$= \frac{\pi}{2} - \theta$$

$$y = \frac{\pi}{2} - \cos^{-1}(u)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dx}}{\frac{dx}{dy}} = \frac{-1}{2}$$

$$x = \sec^{-1}\left(\frac{1}{cos\ 2\theta}\right)$$

$$= \sec^{-1}\sec 2\theta$$

$$x = 2\theta$$

$$u = 2\cos^{-1}(u)$$

$$\frac{dx}{du} = \frac{-2}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dx}}{\frac{dx}{dy}} = \frac{-1}{2}$$

Q18. If $y = \cot^{-1}\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}$, find $\frac{dy}{dx} x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$.

Ans. $\frac{1}{2}$ or $-\frac{1}{2}$

Sol.
$$E = \frac{\sqrt{1\ +\ sin}\ +\ \sqrt{1-sin\ x}}{\sqrt{1+sin\ x}-\sqrt{1-sin\ x}} = \frac{1+\sqrt{\frac{1-sin\ x}{1+sin\ x}}}{1+\sqrt{\frac{1-sin\ x}{1-sin\ x}}}$$

$$E = \frac{1+\left|tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|}{1-\left|tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|}$$

$$x \in \left(0, \frac{\pi}{2}\right)$$

$$E = \frac{1\ +\ tan\left(\frac{\pi}{4}-\frac{x}{2}\right)}{1-tan\left(\frac{\pi}{4}-\frac{x}{2}\right)}$$

$$\tan\left(\frac{\pi}{4}\ + \left(\frac{\pi}{4}-\frac{x}{2}\right)\right.$$

$\cot\frac{x}{2}$

$x \in \left(\frac{\pi}{2}, \pi\right)$

$$E = \frac{1 - tan\left(\frac{\pi}{4} - \frac{x}{2}\right)}{1 + tan\left(\frac{\pi}{4} - \frac{x}{2}\right)} = \tan\left(\frac{\pi}{4} - \left(\frac{\pi}{4} - \frac{x}{2}\right)\right)$$

$$y \begin{cases} \cot^{-1}\cot\frac{x}{2} & x \in \left(0, \frac{\pi}{2}\right) \\ \cot^{-1}\tan\frac{x}{2} & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$$

$$y \begin{cases} \frac{x}{2} & x \in \left(0, \frac{\pi}{2}\right) \\ \frac{\pi}{2}\ \frac{-x}{2} & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$$

$$\frac{dy}{dx} \begin{cases} \frac{1}{2} & x \in \left(0, \frac{\pi}{2}\right) \\ \frac{-1}{2} & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$$

Q19. If $y = \tan^{-1}\frac{x}{1+\sqrt{1-x^2}} + \sin\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right)$, then find $\frac{dy}{dx}$ for $\in (-1, 1)$.

Ans. $\frac{1-2x}{2\sqrt{1-x^2}}$

Sol. $x = \cos\ \theta\ \in (0, \pi)$

$y = \tan^{-1}\left(\frac{cos\ \theta}{1\ +\ sin\ \theta}\right)$

$= \tan^{-1}\tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right)]$

$y = \frac{\pi}{4} - \frac{-\theta}{2}$

$y = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}(x)$

$y = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}(x) + \sqrt{1-x^2}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}}$$

Q20. Let P(x) be a polynomial of degree 4 such that P(1) = P(3) = P(5) = P'(7) = 0. If the real number

$x \neq 1, 3, 5$ is such that P(x) = 0 can be expressed as x = p/q where 'p' and 'q' are relatively prime, then (p + q) equals

Ans. 100

Sol. Let $P(x) = a(x-1)(x-3)(x-3)(x-5)(x-\alpha)$

$P'(7) = 0$

$P'(x) = a$

$$\left((x-3)(x-5)(x-\alpha)+(x-1)(x-5)(x-\alpha)+(x-1)(x-3)(x-\alpha)+(x-1)(x-3)(x-5)\right)$$

Put x = 7

$8(8-\alpha) + 12(7-\alpha) + 12(7-\alpha) + 48 = 0$

$44\alpha = 356$

$$\alpha = \frac{89}{11}$$

Q21.

$$f(x) = \begin{cases} b\sin^{-1}\left(\frac{x+c}{2}\right) & , \quad -\frac{1}{2} < x < 0 \\ \frac{1}{2} & \quad at x = 0 \\ \frac{e^{ax/2}-1}{x} & , \quad 0 < x < \frac{1}{2} \end{cases}.$$

If f(x) is differentiable at x = 0 and | c | < 1/2 then find the value of 'a' and prove that $64b^2 = 4 - c^2$.

Ans. a = 1

Sol. (b) Let $P(x) = ax^2 + bx + c$

$P(0) = 0 \Rightarrow c = 0$

$P(1) = 1 \Rightarrow a + b = 1$

$\therefore \quad P(x) = (1-a)x^2 + ax$

$P'(x) = 2(1-a)x + a > 0$

put $x = 0, \quad a > 0$

$x = 1, \quad a < 2$

$S = \{(1-a)x^2 + ax \,;\, 0 < a < 2\}$.

Q22. If $f(x-y) = f(x)\cdot g(y) - f(y)\cdot g(x)$ and $g(x-y) = g(x)\cdot g(y) + f(x)\cdot f(y)$ for all $x, y \in R$. If right hand derivative at x = 0 exists for $f(x)$. Find derivative of g(x) at x = 0.

Ans. $g'(0) = 0$

Sol. Put y = x

g (0) = $(g(x))^2 + (f(x))^2$

Diff. wrt x

0 = 2g (x) g'(x) + 2f(x) f'(x)

Put x = 0

2g (0) g'(0) + 2f(0) f'(0) = 0

$\therefore$ f(0) = 0 & g(0) $\neq$ = 0

$\Rightarrow$ g'(0) = 0

Q23. Let f(θ) = $\sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)\right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan\theta)}(f(\theta))$

Ans. 1

Sol. Let $f(\theta) = \sin\alpha$ where $\alpha = \tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)$

$$\Rightarrow \tan\alpha = \frac{\sin\theta}{\sqrt{\cos 2\theta}}$$

$$\Rightarrow \sin\alpha = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

$$\Rightarrow f(\theta) = \tan\theta$$

$$\Rightarrow \frac{d(f(\theta))}{d(\tan\theta)} = 1$$

Q24. (a) Let f (x) = $x^2 - 4x - 3$, x > 2 and let g be the inverse of f. Find the value of g ' where f (x) = 2.

(b) Let f : R→R be defined as f (x) = $x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ and g(x) = $f^{-1}(x)$, then find g ' (–1).

(c) Suppose f^{-1} is the inverse function of a differentiable function f and let G(x) = $\frac{1}{f^{-1}(x)}$.

If f(3) = 2 and f '(3) = $\frac{1}{9}$, find G ' (2).

Ans. (a) 1/6, (b) $\frac{1}{14}$ (c) – 1

Sol. (b) g (f (x)) = x

g'(f(x)) f'(x) = 1

Put x = 0

g' (– 1) f'(0) = 1

g' (–1) = $\frac{1}{f'(0)}$

= $\frac{1}{14}$

$f(x) = x^3 + 3x^2 + 6x – 5x + 4e^{2x}$

$f'(x) = 3x^2 + 6x + 6 + 8e^{2x}$

Put x = 0

f'(0) = 6 + 8 = 14

(c) let $g(x) = f^{-1}(x)$

g (f (x)) = x

g'(f(x)) f'(x) = 1

g(x) = $\frac{1}{g(x)}$

Diff. wrt x

g' (x) = $\frac{1}{(g(x))^2}$ g'(x)

Put x = 2

g'(2) = $\frac{1}{(g(2))^2}$ g'(2)

= – $\frac{1}{(3)^2}$(g) = – 1

Q25. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of g'(1) is

Ans. 2

Sol. $f(x) = x^3 + e^{x/2}$, $g(x) = f^{-1}(x)$

⇒ g'(f(x)). f' (x) = 1

Put f(x) = 1 ⇒ $x^3 + e^{x/2} = 1$

⇒ x = 0

Q26. (a) If y = y(x) and it follows the relation $e^{xy} + y \cos x = 2$, then find (i) y ' (0) and (ii) y " (0).

(b) A twice differentiable function f (x) is defined for all real numbers and satisfies the following conditions

f (0) = 2; f ' (0) = – 5 and f " (0) = 3.

The function g (x) is defined by g (x) = eax + f (x) $\forall$ x $\in$ R, where 'a' is any constant. If g ' (0) + g '' (0) = 0.

Find the value(s) of 'a'.

Ans. (a) (i) y ' (0) = – 1 ; (ii) y '' (0) = 2 ; (b) a = 1, – 2

Sol. (a) e^{xy} + y ab x = 2

put x = 0 $\Rightarrow$ y = 1

e^{xy} + y cos x = 2

Diff wrt x

$$e^{xy}\left(y + \frac{xdy}{dx}\right) + \cos x \frac{dy}{dx} - y \sin x = 0$$

Put x = 0 y = 1

$$\Rightarrow \left(\frac{dy}{dx}\right) (g') = -1$$

$$e^{xy} (y + xy_1) + y_1 \cos x - y \sin x = 0$$

Diff wrt x

$$e^{xy} (y + xy_1)^2 + (y_1 + y_1 + xy_2) - y_1 \sin x + y_2 \cos x$$
$$- y_1 \sin x - y \cos x = 0$$

Put x = 0 y = 1 y, = – 1

$\Rightarrow$ $(y_2) = 2$

(b) g (x) = e^{ax} + f(x)

g' (x) = ae^{ax} + f'(x)

g''(x) = $a^2 e^{ax}$ + f'' (x)

g' (0) + g'' (0) = 0

a + f'(0) + a^2 + f''(0) = 0

a – 5 + a^2 + 3 = 0

a^2 + a – 2 = 0

(a + 2) (a – 1) = 0

a = – 2, 1

Q27. If x = 2cost - cos2t and y = 2sint - sin2t, find the value of (d^2y/dx^2) when t = $(\pi/2)$.

Ans. $\dfrac{-3}{2}$

Sol. x = 2cost - cos2t

$$\frac{dx}{dt} = -2\sin t + 2\sin 2t$$

y = 2sint - sin2t

$$\frac{dy}{dt} = 2\cos t - 2\cos 2t$$

$$\frac{dy}{dx} = \frac{2\cos t - 2\cos 2t}{-2\sin t + 2\sin 2t} = \frac{\cos t - \cos 2t}{\sin 2t - \sin t}$$

$$\frac{dy}{dx} = \tan\left(\frac{3t}{2}\right)$$

$$\frac{d^2y}{dx^2} = \frac{3}{2}\sec^2\left(\frac{3t}{2}\right) \times \frac{dt}{dx}$$

$$= \frac{3}{2}\sec^2\left(\frac{3t}{2}\right) \times \frac{1}{2(\sin 2t - \sin t)}$$

$$t = \frac{\pi}{2}$$

$$\frac{d^2y}{dx^2} = \frac{-3}{2}$$

Q28. Find the value of the expression $y^3\frac{d^2y}{dx^2}$ on the ellipse $3x^2 + 4y^2 = 12$.

Ans. $\frac{-9}{4}$

Sol. $3x^2 + 4y^2 = 12$

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

x = $\cos\theta$ y = $\sqrt{3}\sin\theta$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = -\frac{\sqrt{3}}{2}\cot\theta$$

Diff wt x

$$\frac{d^2\,y}{dx^2} = \frac{\sqrt{3}}{2}\,\text{cosec}^2\,\theta\left(\frac{d\,\theta}{dx}\right)$$

$$= \left(\frac{\sqrt{3}}{2}\,cosec^2\,\theta\right)\left(\frac{-1}{2\,sin\,\theta}\right)$$

$$\frac{d^2\,y}{dx^2} = \frac{-\sqrt{3}}{4}\,\text{cosec}^3\,(\theta)$$

$$y^3\,\frac{d^2\,y}{dx^2} = \left(-\frac{\sqrt{3}}{4}cosec^3\,\theta\right)\left(\sqrt{3}\,sin\,\theta\right)^3$$

$= \frac{-9}{4}$

Q29. If f : R → R is a function such that $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3)$ for all $x \in R$, then prove that
$f(2) = f(1) - f(0)$.

Ans. -

Sol. $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$

Put $f'(1) = a$ $\quad f''(2) = b$ $\quad f'''(3)$

$f(x) = x^3 + ax^2 + bx + c$

$f'(x) = 3x^2 + 2ax + b$

$f''(x) = 6x + 2a$

$f'''(x) = 6$

$c = f'''(3) = 6$

$b = f''(2) = 12 + 2a$ ....(1)

$a = b'(1) = 3 + 2a + b$ ....(2)

Solve (1) & (2)

$a = -5$

$b = 2$

$\Rightarrow f(x) = x^3 - 5x^2 + 2x + 6$

Now check

$f(2) = f(1) - f(0)$

Q30. If $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$ then $f'(x) = \lambda$. $\begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$ Find the value of λ.

Ans. 3

Sol.
$$f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$$

$$f'(x) = \begin{vmatrix} 4(x-a)^3 & (x-a)^3 & 1 \\ 4(x-b)^3 & (x-b)^3 & 1 \\ 4(x-c)^3 & (x-c)^3 & 1 \end{vmatrix} + \begin{vmatrix} (x-a)^4 & 3(x-a)^2 & 1 \\ (x-b)^4 & 3(x-b)^2 & 1 \\ (x-c)^4 & 3(x-c)^2 & 1 \end{vmatrix} + \begin{vmatrix} (x-a)^4 & (\\ (x-b)^4 & (\\ (x-c)^4 & (\end{vmatrix}$$

$$f'(x) = 0 + 3\begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix} + 0$$

$\lambda = 3$

Q31. If $f(x) = \begin{vmatrix} \cos(x + x^2) & \sin(x + x^2) & -\cos(x + x^2) \\ \sin(x - x^2) & \cos(x - x^2) & \sin(x - x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$ then find f'(x).

Ans. $2(1 + 2x) . \cos 2(x + x^2)$

Sol.

$$f(x) = \begin{vmatrix} \cos(x + x^2) & \sin(x + x^2) & -\cos(x + x^2) \\ \sin(x - x^2) & \cos(x - x^2) & \sin(x - x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$$

$$f(x) = \sin 2x\left[\cos(x^2 + x - x + x^2)\right] + \sin 2x^2\left[\cos(x + x^2 + x - x^2)\right]$$

$= \sin 2x \cos 2x^2 + \sin 2x^2 \cos 2x$

$f(x) = \sin(2x^2 + 2x)$

$f'(x) = (4x + 2)\cos(2x^2 + 2x)$

$= 2(2x + 1)\cos 2(x + x2)$

Q32. If a be a repeated root of a quadratic equation f(x) = 0 & A(x), B(x), C(x) be the polynomials of degree 3, 4 & 5 respectively, then show that $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\beta) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ is divisible by f(x), where dash denotes the derivative.

Ans. -

Sol. $f(x) = c(x - \alpha)^2$

$$F(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\beta) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

Now $F(\alpha) = 0$ $\therefore$ $(x - \alpha)$ is root of F(x).

$$F'(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\beta) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

Now $F'(\alpha) = 0 \Rightarrow x - \alpha$ is a factor of F'(x). So $(x - \alpha)$ must be repeated at least two times in F(x).

$\Rightarrow$ F(x) is divisible by f(x).

Q33. $\lim_{x\to 0}\left[\frac{1}{x\sin^{-1}x}-\frac{1-x^2}{x^2}\right]$

Ans. $\frac{5}{6}$

Sol. $\lim_{x\to 0}\frac{x-(1-x^2)\ sin^{-1}(x)}{x^2\ sin^{-1}(x)}$

$\lim_{x\to 0}\frac{x-(1-x^2)\ sin^{-1}(x)}{x^2\ (x)}$

L' Hospital

$\lim_{x\to 0}\frac{1-\left(\frac{1-x^2}{\sqrt{1-x^2}}\right)+2x\ sin^{-1}(x)}{3x^2}$

$\lim_{x\to 0}\frac{1-\sqrt{1-x^2}+2x\ sin^{-1}(x)}{3x^2}$

$\lim_{x\to 0}\frac{1-\sqrt{1x^2}}{3x^2}\quad+\lim_{x\to 0}\ \frac{2\ sin^{-1}(x)}{3x}$

$\lim_{x\to 0}\frac{1-(1-x^2)}{3x^2(1+\sqrt{1-x^2})}+\frac{3}{2}$

$\frac{1}{6}+\frac{2}{3}=\frac{5}{6}$

Q34. $\lim_{x \to 0} \frac{x + ln\left(\sqrt{x^2+1} - x\right)}{x^3}$

Ans. $\frac{1}{6}$

Sol. $\lim_{x \to 0} \frac{x \ + \ ln(\sqrt{x^2+1} - x)}{x^3}$

L' Hospital

$$\lim_{x \to 0} \frac{1 \ + \ \left(\frac{1}{\sqrt{x+1}-1}\left(\frac{x}{\sqrt{x^2+1}} - 1\right)\right)}{3x^2}$$

$$\lim_{x \to 0} \frac{1 - \frac{1}{\sqrt{x^2+1}}}{3x^2}$$

$$\lim_{x \to 0} \frac{\sqrt{x^2 \ +1} - 1}{3x^2\sqrt{x^2 \ + \ 1}}$$

$$\lim_{x \to 0} \left(\frac{(x^2 \ + \ 1) - 1}{3x^2\ (\sqrt{x^2 \ + \ 1} - 1)}\right)$$

$$\frac{1}{3\lambda^2} = \frac{1}{6}$$

Q35. $\lim_{x \to 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x}\right]$

Ans. $-\frac{1}{3}$

Sol. $\lim_{x \to 0} \frac{1}{x^2} - \frac{1}{sin^2\ x}$

$$\frac{sin^2\ (x) \ - \ x^2}{x^2\ sin^2\ x}$$

$$\lim_{x \to 0} \frac{(sin\ x - x)(sin\ x \ - x)}{x^3 \quad x}$$

$$\left(\frac{-1}{6}\right)(2) \ = - \ \frac{-1}{3}$$

Q36. (a) $\lim_{x \to ^+} x^{\left(x^x - 1\right)}$

(b) $\lim_{x \to ^+} \left(\cot x - ln^2 x\right)$

Ans. (a) 1 (b) does not exist

Sol. (a) & (b) from picture attached in LMS

Q37. $\lim_{x \to 0} \frac{1 + \sin x - \cos x + ln(1-x)}{x.\tan^2 x}$

Ans. $-\frac{1}{2}$

Sol. $\lim_{x \to 0} \frac{1 + sin\ x\ - cos\ x +\ ln\ (1 - x)}{x^3}$

L' Hospital

$\lim_{x \to 0} \frac{cos\ x\ +\ sin\ x\ -\ \frac{1}{(1-x)^2}}{3x^2}$

$\lim_{x \to 0} \frac{-sin\ x\ +\ cos\ x\ -\ \frac{1}{(1-x)^2}}{6x}$

L' Hospital

$\lim_{x \to 0} \frac{-cos\ x\ +\ sin\ x\ -\ \frac{2}{(1-x)^3}}{6}$

$\frac{-\ 1\ -\ 0\ -\ 2}{6} = \frac{-1}{2}$

Q38. $\lim_{x \to \frac{\pi}{2}} \frac{\sin x - (\sin x)^{\sin x}}{1 - \sin x + ln(\sin x)}$

Ans. 2

Sol. Put sin x = t

$\lim_{t \to t} \frac{t -\ t^t}{1 -\ t\ +\ lnt}$

L' Hospital

$\lim_{t \to t} \frac{1 - t^t\ (1\ +\ ln\ t)}{-\ 1\ +\ \frac{1}{t}}$

$\lim_{t \to t} \frac{-t^t\ (1\ +\ Gt)^2 - t^t \left(\frac{1}{t}\right)}{-\frac{1}{t^2}}$

Put $t = 1$

$\frac{-1 - 1}{-1} = 2$

Q39. If $\lim_{x \to 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 . ln(l + x) - 2x^3 + x^4}$ exists & is finite, find the values of a, b, c & the limit.

Ans. a = 6, b = 6, c = 0; $\frac{3}{40}$

Sol. Use expansion

$$\lim_{x \to 0} \frac{a\left(x - \frac{x^3}{\lfloor 3} + \frac{x^5}{\lfloor 5}\ldots\ldots\right) - bx\ +\ cx^2\ + x^3}{2x^2\left(x\frac{x}{2} + \frac{x^3}{3} - \frac{x^4}{4}\right) - 2x^3\ +\ x^4}$$

$$\lim_{x\to0}\frac{x(a-b)+(x^2+x^3)\left(1-\frac{a}{6}\right)+x^5\left(\frac{a}{20}\right)}{\frac{2}{3}x^5-\frac{2}{4}x^6+\ldots\ldots}$$

$$\left.\begin{array}{r} a-b=0 \\ c=0 \\ 1-\frac{a}{b}=0 \end{array}\right] \quad \Rightarrow \quad \begin{array}{l} a=6 \\ b=0 \\ c=0 \end{array}$$

$$\text{Limit}=\frac{a}{120}\times\frac{3}{2}=\frac{a}{80}=\frac{6}{80}=\frac{3}{40}$$

Q40. Evaluate: $\lim_{x\to0}\frac{x^{6000}-(\sin x)^{6000}}{x^2.(\sin x)^{6000}}$

Ans. 1000

Sol.

$$\lim_{x\to0}\frac{\left(\frac{x}{sin\ x}\right)^{6000}-1}{x^2}$$

$$\lim_{x\to0}\frac{6000\left(\frac{x}{sin\ x}-1\right)}{x^2}$$

$$\lim_{x\to0}\frac{6000\ (x-sin\ x)}{x^2\ \ sin\ x}$$

$$\lim_{x\to0}\frac{6000\ (x-sin\ x)}{x^3}$$

$$6000\ \left(\frac{1}{6}\right)=1000$$

Q41. (a) If $\lim_{x\to0}\frac{1-\cos x\cdot\cos 2x\cdot\cos 3x\ldots..\cos nx}{x^2}$ has the value equal to 253, find the value of n

(where n $\in$ N).

(b) Find the value of $\lim_{x\to0}\frac{1-\cos^5 x\cos^3 2x\cos^3 3x}{x^2}$.

(c) If $\lim_{x\to0}\frac{1-\cos 3x\cdot\cos 9x\cdot\cos 27x\ldots\ldots\ldots.\cos 3^n x}{1-\cos\frac{1}{3}x\cdot\cos\frac{1}{9}x\cdot cos\frac{1}{27}x\ldots\ldots\ldots\cos\frac{1}{3^n}x}=3^{10}$, find the value of n.

Ans. (a) 11, (b) 22, (c) 4

Sol. (a) $\lim_{x\to0}\frac{1-cos\ x\ cos\ x\ 2x\ cos\ 3x\ldots\ldots.cos\ 4x}{x^2}$

$$\lim_{x\to0}\frac{(cos\ x\ cos\ 2x\ cos\ 3x\ldots..cos\ 4x)(tan\ x\ +\ 2tan\ x\ 3\ tan\ 3x+\ldots\ldots4\ tan\ 4x\ )}{x^2}$$

$$\frac{1}{2}\left(\frac{tan\ x}{x}+\frac{2\ tan\ x}{x}+\frac{3\ tan\ 3x}{x}+\ldots\ldots+\frac{4\ tan\ 4x}{x}\right)$$

$$\frac{1}{2}(1^2+2^2+\ldots\ldots\ldots\ldots..+n^2)=253$$

$\Rightarrow$ n = 11

(b) Use result of above part.

$$= \frac{\frac{1}{2}\left((5(1^2)) + 3(2^2) + 3\,(3^2)\right)}{= \frac{1}{5}(5 + 12 + 27) = 12}$$

Paragraph

Let f(x) and g(x) be two differentiable functions, defind as : $f(x)= x^2 + xg\,'(1) + g\,''(2)$ and $g(x) = f(1)\,x^2 + x\,f\,'(x) + f\,''(x)$

\n

Q42. The value of $f(1) + g(-1)$ is

(A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol. $f(x) = x^2 + x\,g'(1) + g''(2)$

$g'(1) = a\ g''(2) = 2$

$f(x) = x^2 + ax + b$

$f'(x) = 2x + a$

$f''(x) = 2$

$f(1)\ a + b + 1$

$g(x) = f(1)\,x^2 + x\,f'(x) + f''(x)$

$g(x) = (a + b + 1)x^2 + x\,(2x + a) + 2$

$g(x) = (a + b + 3)x^2 + ax + 2$

$g'(x) = 2\,(a + b + 3)\,x + a$

$g''(x) = 2\,(a + b + 3)$

$a = 2\,(a + b + 3) + a$ ....(1)

$b = 2\,(a + b + 3)$ ....(2)

On solving (1) & (2)

$a = -3$ & $b = 0$

$\Rightarrow f(x) = x^2 - 3x$

$9x = 2 - 3x$

$f(1) + g(-1) = (1 - 3) + (2 + 3) = 3$

Paragraph

Let f(x) and g(x) be two differentiable functions, defind as : f(x)= x^2 + xg '(1) + g '' (2) and g(x) = f(1) x^2 + x f ' (x) + f ''(x)

\n

Q43. The number of integers in the domain of the function F(x) = $\sqrt{-\frac{f(x)}{g(x)}} + \sqrt{3-x}$ is

(A) 0 (B) 1 (C) 2 (D) infinite

Ans. C

Sol. f(x) = x^2 + x g'(1) + g'' (2)

g'(1) = a g''(2) = 2

f(x) = x^2 + ax + b

f'(x) = 2x + a

f'' (x) = 2

f(1) a + b + 1

g (x) = f(1) x^2 + x f'(x) + f''(x)

g(x) = (a + b + 1)x^2 + x (2x + a) + 2

g (x) = (a + b + 3)x^2 + ax + 2

g' (x) = 2 (a + b + 3) x + a

g'' (x) = 2 (a + b + 3)

a = 2 (a + b + 3) + a ....(1)

b = 2 (a + b + 3) ....(2)

On solving (1) & (2)

a = – 3 & b = 0

$\Rightarrow$ f(x) = x^2 – 3x

9x = 2 – 3x

f(x) = $\sqrt{-\frac{f(x)}{g(x)}}$ $+\sqrt{3-x}$

$-\frac{f(x)}{g(x)} \geq 0$ $3 - x \geq 0$

$x \leq 0$

$-\frac{(x^2 - 3x)}{2-3x} \geq 0$

$$\frac{(x-3x)}{3x-2x} \geq 0$$

$$x \left[0, \frac{2}{3}\right) U \; \{3\}$$

Integers in domain = {0, 3}

Q44. If $f(x) = x\,.\,|x|$, then its derivative is :

(A) $2x$ (B) $-2x$ (C) $2|x|$ (D) $2x \text{ sgn } x$

Ans. D, C

Sol.

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 2x & x \geq 0 \\ -2x & x < 0 \end{cases}$$

Exercise JA

Q1. Let f(x) = x sin px, x > 0. Then for all natural numbers n, f'(x) vanishes at

(A) a unique point in the interval $\left(n, n+\frac{1}{2}\right)$

(B) a unique point in the interval $\left(n+\frac{1}{2}, n+1\right)$

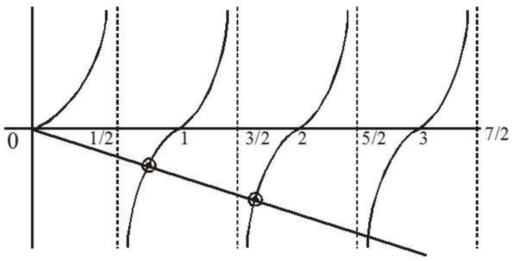
(C) a unique point in the interval (n, n + 1) (D) two points in the interval (n, n + 1)

Ans. B, C

Sol. l' (x) = sin πx + πx cos πx = 0

tan πx = – πx

y = tan πx & y = – πx



intersection point lies in

$\left(\frac{1}{2}, 1\right) \cup \left(\frac{3}{2}, 2\right) \cup \left(\frac{5}{2}, 3\right)$.......

option (B) is correct for $\left(n + \frac{1}{2}, n\right)$

as weel (n, (n + 1)) because root lies in

(0, 1) $\cup$ (1, 2) $\cup$ (2, 3)

Q2. Let f : R → R, g : R→ R and h : R→R be differentiable functions such that f(x) = x^3 + 3x + 2, g(f(x)) = x and h(g(g(x))) = x for all x $\in$ R. Then

(A) g'(2) =$\frac{1}{15}$ (B) h'(1) = 666 (C) h(0) = 16 (D) h(g(3)) = 36

Ans. C, B

Sol. Here, g(f(x)) = x ⇒ g(x) = f^{-1}(x)

h(g(g(x))) = x ⇒ h(g(x)) = f(x) (i)

h(g(g(x))) = x ⇒ h(x) = f(f(x)) (ii)

(A) f(g(x)) = x ⇒ f ' (g(x)) g'(x) = 1 ∴ g'(2) = $\frac{1}{f'(g(2))} = \frac{1}{f'(0)} = \frac{1}{3}$

(B) h'(x) = f '(f(x)) . f '(x)   ($\therefore$ using (ii))

Put x = 1 $\Rightarrow$ h '(1) = 666

(C) h(0) = f(f(0)) = 16

(D) h(g(3)) = f(3)   ($\therefore$ using (i))

Q3. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let f(x) = sin (π cos x) and g(x) = cos (2π sin x) be two functions defined for x > 0.

Define the following sets whose elements are written in the increasing order :

X = {x : f(x) = 0},   Y = {x : f '(x) = 0}

Z = {x : g(x) = 0},   W = {x : g '(x) = 0}

List-I contains the sets X, Y, Z and W. List II contains some information regarding these sets.

List I	List II
(I) x	(P) $\supseteq \left\{\frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi\right\}$
(II) Y	(Q) an arithmetic progression
(III) Z	(R) NOT an arithmetic progression
(IV) W	(s) $\supseteq \left\{\frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}\right\}$
	(T) $\supseteq \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi\right\}$
	(U) $\supseteq \left\{\frac{\pi}{6}, \frac{3\pi}{4}\right\}$

Which of the following is the only CORRECT combination :

(A) (I), (Q), (U)   (B) (II), (R), (S)   (C) (I), (P), (R)   (D) (II), (Q), (T)

Ans. 5f89210fcc30e0082f87a0df

Sol. X :   f(x) = 0

sin (π cosx) = 0

π cosx = nπ n $\in$ I

cox = n   n $\in$ I

$\therefore \in \left\{n\pi, \left(n+\frac{1}{2}\right)\pi\right\}$

$\therefore$ x = $\left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}, \ldots\ldots\right\}$

Y: f'(x) = 0

cos(πcosx). π(–sin x) = 0

cos (π cosx) = 0   or sin x = 0

π cosx = (2n+1) $\frac{\pi}{2}$ x = nπ

cox x = $\frac{(2n+1)}{2}$

$\therefore \cos x = \frac{1}{2}, \frac{-1}{2}$

$x = 2n\pi \pm \frac{\pi}{3} = \left(2n + \frac{1}{3}\right)\pi, \left(2n\frac{-1}{3}\right)\pi$

$x = 2n\pi \pm \frac{2\pi}{3} = \left(2n + \frac{2}{3}\right)\pi, \left(2n\frac{-2}{3}\right)\pi$

$y = \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \ldots\right\}$

Z: g(x) = 0

$\cos(2\pi \sin x) = 0$

$2\pi \sin x = (2n+1)p/2 \quad n \in I$

$\sin x = \frac{(2n+1)}{4}$

$\sin x = \frac{1}{4}, \frac{-1}{4}, \frac{3}{4}, \frac{-3}{4}$

$x = n\pi \pm (-1)^n \sin^{-1}(1/4)$

or $x = n\pi \pm (-1)^n \sin^{-1}(3/4)$

W: g'(x) = 0

$-\sin(2\pi \sin x). \quad 2\pi \cos x = 0$

$\sin(2\pi \sin x) = 0$ or $\cos x = 0$

$2\pi \sin x = n\pi$ $\qquad x = (2n+1)\frac{\pi}{2}$

$\sin x = n/2$

$\therefore \sin x = 0, 1/2, \frac{-1}{2}, 1, -1$

$\therefore x = n\pi$

or $x = n\pi \pm (-1)^n \frac{\pi}{6}$

or $x = 2n\pi \pm \frac{\pi}{2}$

Q4. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$.

Define the following sets whose elements are written in the increasing order :

$X = \{x : f(x) = 0\}, \quad Y = \{x : f'(x) = 0\}$

$Z = \{x : g(x) = 0\}, \quad W = \{x : g'(x) = 0\}$

List-I contains the sets X, Y, Z and W. List II contains some information regarding these sets.

List I | List II

(I) X (P) $\supseteq \left\{\frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi\right\}$

(II) Y (Q) an arithmetic progression

(III) Z (R) NOT an arithmetic progression

(IV) W (s) $\supseteq \left\{\frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}\right\}$

(T) $\supseteq \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi\right\}$

(U) $\supseteq \left\{\frac{\pi}{6}, \frac{3\pi}{4}\right\}$

Which of the following is the only CORRECT combination :

(A) (III), (P), (Q), (U) (B) (III), (R), (U) (C) (IV), (P), (R), (S)

(D) (IV), (Q), (T)

Ans. 5f892347cc30e0082f87a118

Sol. X : f(x) = 0

$\sin(\pi \cos x) = 0$

$\pi \cos x = n\pi \quad n \in I$

$\cos x = n \quad n \in I$

$\therefore \in \left\{n\pi, \left(n+\frac{1}{2}\right)\pi\right\}$

$\therefore x = \left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}, \ldots\ldots\right\}$

Y: f'(x) = 0

$\cos(\pi\cos x).\ \pi(-\sin x) = 0$

$\cos(\pi \cos x) = 0$ or $\sin x = 0$

$\pi \cos x = (2n+1)\frac{\pi}{2} \qquad x = n\pi$

$\cos x = \frac{(2n+1)}{2}$

$\therefore \cos x = \frac{1}{2}, \frac{-1}{2}$

$x = 2n\pi \pm \frac{\pi}{3} = \left(2n+\frac{1}{3}\right)\pi, \left(2n\frac{-1}{3}\right)\pi$

$x = 2n\pi \pm \frac{2\pi}{3} = \left(2n+\frac{2}{3}\right)\pi, \left(2n\frac{-2}{3}\right)\pi$

$y = \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \ldots\right\}$

Z: g(x) = 0

$\cos(2\pi \sin x) = 0$

$2\pi \sin x = (2n+1)p/2 \quad n \in I$

$\sin x = \frac{(2n+1)}{4}$

$\sin x = \frac{1}{4}, \frac{-1}{4}, \frac{3}{4}, \frac{-3}{4}$

$x = n\pi \pm (-1)^n \sin^{-1}(1/4)$

or $x = n\pi \pm (-1)^n \sin^{-1}(3/4)$

W: $g'(x) = 0$

$-\sin(2\pi \sin x). \quad 2\pi \cos x = 0$

$\sin(2\pi \sin x) = 0$ or $\cos x = 0$

$2\pi \sin x = n\pi$ $\qquad\qquad x = (2n+1)\frac{\pi}{2}$

$\sin x = n/2$

$\therefore \sin x = 0, 1/2, \frac{-1}{2}, 1, -1$

$\therefore x = n\pi$

or $x = n\pi \pm (-1)^n \frac{\pi}{6}$

or $x = 2n\pi \pm \frac{\pi}{2}$

Q5. Let b be a nonzero real number. Suppose $f: R \to R$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$ for all $x \in R$, then which of the following statements is/are TRUE?

(A) If $b > 0$, then f is an increasing function

(B) If $b < 0$, then f is a decreasing function

(C) $f(x) f(-x) = 1$ for all $x \in R$

(D) $f(x) - f(-x) = 0$ for all $x \in R$

Ans. A, C

Sol. $f'(x) = \frac{f(x)}{b^2 + x^2}$

$\int \frac{f'(x)}{f(x)} dx = \int \frac{dx}{x^2 + b^2}$

$\Rightarrow \ln |f(x)| = \frac{1}{b} \tan^{-1}\left(\frac{x}{b}\right) + c$

Now $f(0) = 1$

$\therefore c = 0$

$\therefore |f(x)| = e^{\frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right)}$

$\Rightarrow f(x) = \pm e^{\frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right)}$

since $f(0) = 1$ $\qquad \therefore f(x) = e^{\frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right)}$$

$x \to -x$

$f(-x) = e^{-\frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right)}$

$\therefore f(x) \cdot f(-x) = e^{0} = 1$ (option C)

and for $b > 0$

$f(x) = e^{\frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right)}$

$\Rightarrow f(x)$ is increasing for all $x \in R$ (option A)

Q6. Let α be a positive real number. Let $f : R \to R$ and g : (α, ∞) → R be the functions defined by $f(x) = \sin\left(\frac{\pi x}{12}\right)$ and $g(x) = \dfrac{2\log_e\left(\sqrt{x} - \sqrt{\alpha}\right)}{\log_e\left(e^{\sqrt{x}} - e^{\sqrt{\alpha}}\right)}$. Then the value of $\lim\limits_{x \to \alpha^+} f(g(x))$ is ______.

Ans. 00.50

Sol.

$$\lim_{x \to \alpha^+} g(x) = \lim_{x \to \alpha^+} \frac{\frac{2}{\sqrt{x}-\sqrt{\alpha}}\left(\frac{1}{2\sqrt{x}}\right)}{\frac{1}{e^{\sqrt{x}}-e^{\sqrt{\alpha}}}\left(\frac{1}{2\sqrt{x}} \cdot e^{\sqrt{x}}\right)} \quad \text{(Using L'Hospital Rule)}$$

$$= \lim_{x \to \alpha^+} \frac{e^{\sqrt{x}} - e^{\sqrt{\alpha}}}{\sqrt{x} - \sqrt{\alpha}} \cdot \frac{1}{e^{\sqrt{x}}} \cdot 2$$

$$= \lim_{x \to \alpha^+} \frac{e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}}{\frac{1}{2\sqrt{x}}} \cdot \frac{2}{e^{\sqrt{x}}} = 2$$

$$\lim_{x \to \alpha^+} f(g(x)) = f\left(\lim_{x \to \alpha^+} g(x)\right) = \sin\frac{\pi}{6} = \frac{1}{2} = 00.50$$

Exercise C1

Q1. If y = (x + $\sqrt{1+x^2}$)^n then (1+ x^2) y_2+ xy^1 =

(A) ny^2 (B) n^2y (C) n^2y^2 (D) None of these

Ans. B

Sol.
$$y = \left(x + \sqrt{1+x^2}\right)^n$$

$$y_1 = n\left(x + \sqrt{1+x^2}\right)^{n-1}\left(1 + \frac{2x}{2\sqrt{1+x^2}}\right)$$

$$y_1 = \frac{n\left(x + \sqrt{1+x^2}\right)^n}{\sqrt{1+x^2}}$$

$$y_2 = \frac{n^2\left(x + \sqrt{1+x^2}\right)^{n-1}\left(1 + \frac{x}{\sqrt{1+x^2}}\right)\sqrt{1+x^2} - n\left(x + \sqrt{1+x^2}\right)^n \frac{2x}{2\sqrt{1+x^2}}}{(1+x^2)}$$

$$(1+x^2)y_2 = n^2\left(x + \sqrt{1+x^2}\right)^n - \frac{n\left(x + \sqrt{1+x^2}\right)^n}{\sqrt{1+x^2}}$$

$(1 + x^2)y_2 = n^2y – y_1x$

$(1 + x^2)y^2 + xy_1 = n^2y$

Q2. Let u(x) and v(x) are differentiable functions such that $\dfrac{u(x)}{v(x)} = 7$. If $\dfrac{u'(x)}{v'(x)} = p$ and $\left(\dfrac{u(x)}{v(x)}\right)' = q$, then $\dfrac{p+q}{p-q}$ has the value equal to

(A) 1 (B) 0 (C) 7 (D) -7

Ans. A

Sol.
$$\frac{u(x)}{v(x)} = 7 \qquad \Rightarrow \left(\frac{u(x)}{v(x)}\right)' = 0 = q$$

$$u(x) = 7v(x)$$

$$\Rightarrow u'(x) = 7v'(x) \qquad \frac{p+q}{p-q} = \frac{7+0}{7-0} = 1$$

$$\Rightarrow \frac{u'(x)}{v'(x)} = 7 = p$$

Q3. If $x = t^3 + t + 5$ & $y = \sin t$ then $\dfrac{d^y}{dx^2}$ =

(A) $-\dfrac{(3t^2+1)\sin t + 6t\cos t}{(3t^2+1)^3}$ (B) $\dfrac{(3t^2+1)\sin t + 6t\cos t}{(3t^2+1)^3}$ (C) $-\dfrac{(3t^2+1)\sin t + 6t\cos t}{(3t^2+1)^2}$

(D) $\dfrac{\cos t}{3t^2+1}$

Ans. A

Sol. -

Q4. If $f(x) = \dfrac{a+\sqrt{a^2-x^2}+x}{\sqrt{a^2-x^2}+a-x}$ where a > 0 and x < a, then f ' (0) has the value equal to

(A) $\sqrt{a}$ (B) a (C) $\dfrac{1}{\sqrt{a}}$ (D) $\dfrac{1}{a}$

Ans. D

Sol. $$y = \frac{(a\ +\ x)\ +\sqrt{a-x}.\sqrt{a+x}}{(a\ -x)+\sqrt{a-x}\sqrt{a+x}}$$

$$y = \frac{\sqrt{a\ +\ x}\left(\sqrt{a\ +\ x}\ +\sqrt{a\ +\ x}\sqrt{a\ -x}\right)}{\sqrt{a\ -\ x}\left(\sqrt{a\ +\ x}\ +\sqrt{a\ +\ x}\sqrt{a\ -x}\right)} = \left(\frac{a\ +\ x}{a\ +\ x}\right)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2}\sqrt{\frac{a\ -x}{a\ +x}}\left(\frac{(a\ -\ x)\ +\ (a\ +\ x)}{(a\ -\ x)^2}\right)\frac{1}{2}\sqrt{\frac{a\ -x}{a\ +x}}\times\frac{2a}{(a-x)};\quad \therefore\quad \left.\frac{dy}{dx}\right|_{x=0} = \frac{1}{a}\text{ Ans.}$$

Q5. Suppose that f (0) = 0 and f ' (0) = 2, and let g (x) = . The value of g ' (0) is equal to

(A) 0 (B) 1 (C) 6 (D) 8

Ans. C

Sol. $g(x) = \left(-x+f\left(f(x)\right)\right)$; f (0) = 0; f ' (0) = 2

$$g'(x) = \left(-x+f\left(f(x)\right)\right).\left[-1+f'\left(f(x)\right).f'(x)\right]$$

$$g'(0) = f'\left(f(0)\right).\left[-1+f'(0).f'(0)\right]$$

$$= f'(0)\,[-1+(2)(2)]$$

$$= (2)\,(3) = 6$$

Q6. If $\ln(3\sin x - 4\cos x + 7 + 5y) = (\sin^2 x)y$, then $y'(\pi)$ is equal to

(A) $\dfrac{5}{3}$ (B) 0 (C) $\dfrac{3}{5}$ (D) $\dfrac{-3}{5}$

Ans. C

Sol. We have $\ln(3\sin x - 4\cos x + 7 + 5y) =$ .... (1)

Put $x = \pi$ in equation (1), we get $\ln(11+5y) = 0 \Rightarrow 11 + 5y = 1 \Rightarrow 5y = -10$

$\Rightarrow y = -2$. So, $(\pi, -2)$ lies on the given curve.

Now on differentiating both the sides of equation (1) w.r.t. x, we get

$$\frac{1}{(3\,sin\,x - 4\,cos\,x\ +\ 7\,5y)}\times\left(3\,cos\,x\ +\ 4\,sin\,x + 5\frac{dy}{dx}\right) = (\sin^2 x)\,\frac{dy}{dx} + (\sin 2x)y$$

as $(\pi, -2)$ satisfy it, we get $1 \times \left(-3 + 0\,5\frac{dy}{dx}\right) = 0$

$\therefore \left.\frac{dy}{dx}\right]_{(\pi,-2)} = \frac{3}{5} \Rightarrow y'(\pi)\ \frac{3}{5}$

Q7. If $f''(x) = -f(x)$ and $g(x) = f'(x)$ and $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and given that $F(5) = 5$, then $F(10)$ is equal to :

(A) 5 (B) 10 (C) 0 (D) 15

Ans. A

Sol. $g(x) = f'(x)$ ..... (i)

$\Rightarrow g'(x) = f''(x) = -f(x)$ ..... (ii)

$$F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$$

$\Rightarrow F'(x) = 2f(x/2)\,.\,f'(x/2)\,.\ + 2\,g(x/2)\,.\,g¢(x/2)\,.\frac{1}{2}$

$$\Rightarrow F'(x) = f\left(\frac{x}{2}\right)f'\left(\frac{x}{2}\right) + g\left(\frac{x}{2}\right)g'\left(\frac{x}{2}\right)$$

$$\Rightarrow F'(x) = f\left(\frac{x}{2}\right)\cdot f'\left(\frac{x}{2}\right) - f'\left(\frac{x}{2}\right)f\left(\frac{x}{2}\right) \quad \{\text{Using (i) \& (ii)}\}$$

$\Rightarrow F'(x) = 0$

$\Rightarrow F(x) = \text{constant} \Rightarrow F(5) = F(10) = 5$

Q8. If $y = (\tan^{-1}x)^2$ then $(x^2 + 1)^2 D^2y + 2x(x^2 + 1)\,Dy$ has the value equal to

(A) 0 (B) 2 (C) 4 (D) none

Ans. B

Sol. $(1 + x^2)\,y_1 = 2\tan^{-1}x$

$(1 + x^2)^2 \cdot y_1^2 = 4(\tan^{-1}x)^2 = 4y$

$(1 + x^2)^2 \cdot 2y_1y_2 + y_1^2 \cdot 2(1 + x^2)\cdot 2x = 4y_1$

$(1 + x^2)^2\,y_2 + 2x(1 + x^2)\,y_1 = 2$ Ans.]

Alternatively: $y = (\tan^{-1}x)^2$

$y_1 = \frac{2\tan^{-1}x}{1 + x^2}$; $(1 + x^2)\,y_1 = 2\tan^{-1}x$; $(1 + x^2)y_2 + 2xy_1 = \frac{2}{1 + x^2}$

$(1 + x^2)^2y_2 + 2x(1 + x^2)y_1 = 2$

Hence $k = 2$ Ans.

Q9. Let $g(x)$ be an antiderivative for $f(x)$. Then $\ln(1 + (g(x))^2)$ is an antiderivative for

(A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) none

Ans. B

Sol. Given $\int f(x)\,dx = g(x) \Rightarrow g'(x) = f(x)$

now $\frac{d}{dx}\left(ln\,(1 + g^2(x))\right) = \frac{2g(x)g'(x)}{1 + g^2(x)} = \frac{2f(x)g'(x)}{1 + g^2(x)} \Rightarrow$ (B)

Q10. If $y = (A + Bx)\,e^{mx} + (m-1)^{-2}\,e^x$ then $\frac{d^y}{dx^2} - 2m\frac{dy}{dx} + m^2y$ is equal to :

(A) e^x (B) e^{mx} (C) e^{-mx} (D) $e^{(1-m)x}$

Ans. A

Sol.

$y = (A + Bx)e^{mx} + (m-1)^{-2} \cdot e^x$

$y \cdot e^{-mx} = (A + Bx) + (m-1)^{-2} \cdot e^{(1-x)x}$

$e^{-mx} \cdot y_1 - my + e^{-mx} = B - (m-1)^{-1} \cdot e^{-(m-1)x}$

$e^{-mx} \cdot y_2 - y_1\,e^{-mx} \cdot m - m[e^{-mx} \cdot y_1 - y\,e^{-mx} \cdot m] = e^{-(m-1)x}$

$e^{-mx} \cdot y_1 - m_2y_1\,e^{-mx} + my \cdot e^{-mx} = e^{-(m-1)x}$

$y_2 - 2my_1 + my = e^x$ Ans.

Q11. Let $f(x) = x^n$, n being a non-negative integer. The number of values of n for which $f'(p + q) = f'(p) + f'(q)$ is valid for all $p, q > 0$ is

(A) 0 (B) 1 (C) 2 (D) none of these

Ans. C

Sol. $n = 2$ or 0 only

Q12. The ends A and B of a rod of length $\sqrt{5}$ are sliding along the curve $y = 2x^2$. Let x_A and x_B be the x-coordinate of the ends. At the moment when A is at (0, 0) and B is at (1, 2) the derivative $\frac{dx_B}{dx_A}$ has the value equal to

(A) 1/3 (B) 1/5 (C) 1/8 (D) 1/9

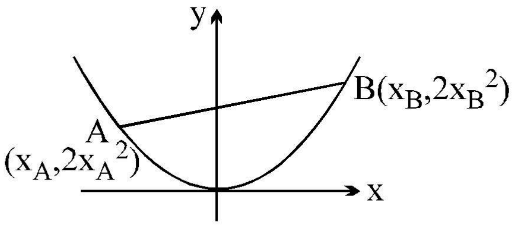
Ans. D

Sol. We have $y = 2x^2$

$(AB)^2 = (X_B - X_A)^2 + (2x_B^2 - 2x_A^2)^2 = 5$

or $(X_B - X_A)^2 + 4\,(2x_B - x_A^2)^2 = 5$

differentiating w.r.t. xA and denoting $\frac{dx_B}{dx_A} = D$



$2(x_B - x_A)(D - 1) + 8\ (2x_B^2\ - x_A^2)(2x_B D - 2x_A) = 0$

put $x_A = 0$; $x_B = 1$

$2(1 - 0)(D - 1) + 8(1 - 0)(2D - 0) = 0$

$2D - 2 + 16D = 0 \Rightarrow D = 1/9$ Ans.

Q13. The derivative of the function,
$f(x)=\cos^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x - 3\sin x)\right\} + \sin^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x - 3\sin x)\right\}$ w.r.t. $\sqrt{1+x^2}$ at $x = \frac{3}{4}$ is

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{10}{3}$ (D) 0

Ans. C

Sol. Put $\cos\phi = \frac{2}{\sqrt{13}}$; $\sin\phi$; $= \frac{3}{\sqrt{13}}$ $\tan\phi = \frac{3}{2}$

$y \cos^{-1}\{\cos(x + \phi) + \sin^{-1}\{\cos(x - \phi)\}$

$= \cos^{-1}\{\cos(x + \phi) + \frac{\pi}{2} - \cos^{-1}\{\cos(\phi - x)\}$ (think !)

$= x + \phi + \frac{\pi}{2} - \phi +$

$y = 2x + \frac{\pi}{2}$; $z = \sqrt{1\ +\ x^2}$

now compute $\frac{dy}{dz}$

Q14. Let $f(x) = \sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right) + \sin^3$ then the primitive of $f(x)$ w.r.t. x is

where C is an arbitrary constant.

(A) $-\frac{3\sin 3x}{4} + c$ (B) $-\frac{3\cos 3x}{4} + c$ (C) $\frac{\sin 3x}{4} + c$ (D) $\frac{\cos 3x}{4} + c$

Ans. D

Sol. Note that $\sin x + \sin\left(x + \frac{2\pi}{3}\right) + \sin\left(x + \frac{4\pi}{3}\right) = 0$

$\Rightarrow \sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right) = -\frac{3}{4}\sin 3x$ (a + b + c = 0
$a^3 + b^3 + c^3 = abc$)

$\therefore \frac{3}{4}\int$ sin 3x dx = $\frac{cos\ 3x}{4}$ + C

Q15. People living at Mars, instead of the usual definition of derivative D f(x), define a new kind of derivative, Df(x) by the formula

Df(x) = $\lim\limits_{h=\to 0} \frac{f^2(x+h) - f^2(x)}{h}$ where f^2 (x) means $[f(x)]^2$. If f(x) = x lnx then D*f(X) $|_{x = e}$ has the value

(A) e (B) 2e (C) 4e (D) 8e

Ans. C

Sol. Hint: D*f(x) = 2f(x).f ¢ (x)

D*(x lnx) = 2x lnx (1 + lnx)]

Q16. Let P(x) be the polynomial $x^3 + ax^2 + bx + c$, where a, b, c $\in$ R. If P(–3) = P(+2) = 0 and P'(–3) < 0, which of the following is a possible value of 'c'?

(A) – 27 (B) – 18 (C) – 6 (D) – 3

Ans. A

Sol. $P(x) = x^3 + ax^2 + bx + c$

$P'(x) = 3x^2 + 2ax + b$

$P(-3) = 0 \Rightarrow -27 + 9a - 3b + c = 0$ ....(1)

$P(2) = 0 \Rightarrow 8 + 4a + 2b + c = 0$ ....(2)

hence (1) – (2) gives

$5a - 5b - 35 = 0$

$a - b = 7$ ....(3)

$P'(-3) = 27 - 6a + b < 0 \Rightarrow 27 - 6(a - b) - 5b < 0 \Rightarrow 27 - 42 - 5b < 0$

$\Rightarrow -15 - 5b < 0$ Þ $3 + b > 0 \Rightarrow b > -3$, hence $a - 7 = b > -3$

$a - 7 > -3$ [from (3)]

$a > 4$ ....(4)

$\therefore$ from (2)

$8 + 16 - 6 + c_{max} = 0$

$c_{max} = -18 \Rightarrow c < -18$

Aliter: $P(x) \equiv (x + 3)(x - 2)(x - a)$

where a is the third root i.e. $P(x) = x^3 + (1 - \alpha)x^2 + (-6 - \alpha)x + 6\alpha$ now proceed.

Q17. f is a differentiable function such that $x = f(t^2)$, $y = f(t^3)$ and $f'(1) \neq 0$ $\left(\frac{d^y}{dx^2}\right)_{t=1}$ =

(A) $\frac{3}{4}\left(\frac{f''(1) + f'(1)}{(f'(1))^2}\right)$ (B) $\frac{3}{4}\left(\frac{f'(1).f''(1) - f''(1)}{(f'(1))^2}\right)$ (C) $\frac{4}{3}\frac{f''(1)}{(f'(1))^2}$

(D) $\frac{4}{3}\left(\frac{f'(1).f''(1)-f''(1)}{(f'(1))^2}\right)$

Ans. A

Sol. $x = f(t^2)$ $\quad\quad\quad\quad$ $y = f(t^3)$

$\frac{dx}{dt} = 2t\, f'(t^2)$ $\quad\quad\quad\quad$ $= 3t^2\, f'(t^2)$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t\, f'(t^3)}{2f'(t^2)}$$

Diff. wrt x

$$\frac{d^2}{dx^2} = \frac{3}{2}\left(\frac{f'(t^2)(f'(t^3)\ +\ 3t^2\ f''(t^3) - tf'(t^3)(2tf''(t^2))}{(f'(t^2))^2}\right)$$

Put t = 1 $\frac{dt}{dx} = \frac{1}{2f'(1)}$

$$= \frac{3}{2}\left(\frac{f'(1)\Big(f'(1)\ + 3f'(1)\Big) - f'(1)\Big(2f''(1)\Big)}{\Big(f'(1)\Big)}\right)\frac{1}{2f(1)}$$

$$= \frac{\Big(f'(1)\ +\ f''(1)\Big)}{4\Big(f'(1)\Big)^2}$$

Q18. If P(x) is a polynomial of degree less than or equal to 2 and S is the set of all such polynomials so that

$P(1) = 1$, $P(0) = 0$ and $P'(x) > 0\ \forall\ x \in [0, 1]$, then

(A) $S = \phi$ (B) $S = \{(1-a)x^2 + ax,\ 0 < a < 2\}$ (C) $S=\{(1-a)x^2 + ax,\ a \in (0, \infty)\}$

(D) $S = \{(1-a)x^2 + ax,\ 0 < a < 1\}$

Ans. B

Sol. Let $P(x) = bx^2 + ax + c$ $\quad\quad$ $b \neq 0$

$P(0) = 0 \quad \Rightarrow \quad c = 0$

$P(1) = 1 \quad \Rightarrow \quad a + b = 1$

$P(x) = (1-a)\,x^2 + ax$

$P'(x) = 2(1-a)x + a$

Now $\quad P'(x) > 0 \quad$ for $\quad x \in [0, 1]$

$P'(0) > 0 \quad \& \quad P'(1) > 0$

$\Rightarrow \quad a > 0 \quad \& \quad 2(1-a) + a > 0$

$\Rightarrow \quad 0 < a < 2$

$\Rightarrow \quad S = \{(1-a)\,x^2 + ax;\ 0 < a < 2\}$

Q19. If $f(x)$ is a continuous and differentiable function and $f(1/n) = 0$, $\forall\ n \geq 1$ and $n \in I$, then

(A) $f(x) = 0,\ x \in (0, 1]$ (B) $f(0) = 0,\ f'(0) = 0$ (C) $f'(x) = 0 = f''(x),\ x \in (0, 1]$

(D) $f(0) = 0$ and $f'(0)$ need not to be zero

Ans. B

Sol. $f(0) = \lim\limits_{x \to \infty} f\left(\frac{1}{n}\right) = 0$

$$f'(0) = \lim_{x \to \infty} \frac{f\left(\frac{1}{n}\right) - f(0)}{\frac{1}{n}}$$

$$\lim_{x \to \infty} = \frac{0 - 0}{\frac{1}{n}} = 0$$

Q20. For the function $y = f(x) = (x^2 + bx + c)e^x$, which of the following holds?

(A) if $f(x) > 0$ for all real x f $\not\Rightarrow$ $(x) > 0$

(B) if $f(x) > 0$ for all real x $\Rightarrow$ $f'(x) > 0$

(C) if $f'(x) > 0$ for all real x $\Rightarrow$ $f(x) > 0$

(D) if $f'(x) > 0$ for all real x $\not\Rightarrow$ $f(x) > 0$

Ans. C, A

Sol. $f(x) = (x^2 + bx + c)e^x$

$\therefore f'(x) = (x^2 + b(b + 2)x + b(b + c)e^x$

$f(x) > 0$ if f $D = b^2 - 4c < 0$

now $f'(x) > 0$ if f $D' = (b + 2)^2 - 4(b - c) = D + 4 < 0$

Thus for $f'(x) >$ 0 $D + 4 < 0$ holds. $\Rightarrow$ $D < 0$ $\Rightarrow$ $f(x) < 0$

Note that the converse need not be true, e.g. $b = c = 1$, $f(x) > 0$ but $f'(-1) = 0$]

Q21. If $y = \tan x\ \tan 2x\ \tan 3x$ then $\frac{dy}{dx}$ has the value equal to

(A) $3\sec^2 3x\ \tan x\ \tan 2x + \sec^2 x\ \tan 2x\ \tan 3x + 2\sec^2 2x\ \tan 3x\ \tan x$

(B) $2y(\operatorname{cosec} 2x + 2\operatorname{cosec} 4x + 3\operatorname{cosec} 6x)$ (C) $3\sec^2 3x - 2\sec^2 2x - \sec^2 x$

(D) $\sec^2 x + 2\sec^2 2x + 3\sec^2 3x$

Ans. C, B, A

Sol. (A) $y = \tan x \tan 2x \tan 3x$

Diff. wrt x

y = sec^2 x tan 2x tan 3x t 2 sec2 2x tan x tan 3x

+ 3 sec^2 (3x) tan x tan 2x

(B) y = tan x tan 2x tan 3x

Take logarithm

log y = log tan x + log tan 2x + lot tan 3x

Diff.

$$\frac{1}{y}\ \frac{dy}{dx} = \frac{sec^2\ x}{tan\ x} + \frac{2sec^2\ (2x)}{tan\ 2x} + \frac{3\ sec^2\ (x)}{tan\ 3x}$$

$$\frac{1}{4}\ \frac{dy}{dx} = \frac{2}{sin\ 2x} + \frac{4}{sin\ 4x} + \frac{6}{sin\ 6x}$$

$\frac{dy}{dx}$ = 2y (cosec 2x + 2 cosec 4x + 2 cosec 6x)

(C) y = tan x tan 2x tan 3x

3x = 2x + x

tan 3x = $\frac{tan\ 2x + tan\ x}{1 -\ tan\ 2x\ tan\ x}$

tan 3x tan 2x tan x = tan 3x – tan 2x – tan x

y = tan 3x – tan 2x – tan x

Diff. wrt x

$\frac{dy}{dx}$ = 3sec^2 (3x) –2sec^2 (2x) – sec^2 x

Q22. Let f(x) = (x^2 – 3x + 2) (x^2 + 3x + 2) and a, b, g satisfy $\alpha < \beta < \gamma$ are the roots of f '(x) = 0 then which of the following is/are correct ([.] denotes greatest integer function) ?

(A) $[\alpha] = -2$ (B) $[\beta] = -1$ (C) $[\beta] = 0$ (D) $[\alpha] = 1$

Ans. A, C

Sol. f(x) = (x – 1) (x – 2) (x + 1) (x + 2)

= (x^2 – 1) (x^2 –4)

f(x) = x^4 – 5x^2 + 4

f' (x) = 4x^3 – 10x

= 2x (2x^2 – 5)

= 4x $\left(x\ -\ \frac{\sqrt{5}}{2}\right)\ \left(x\ +\frac{\sqrt{5}}{2}\right)$

f'(x) = 0

$\Rightarrow x = 0\ \sqrt{\frac{5}{2}}, -\sqrt{\frac{5}{2}}$

$\alpha = -\sqrt{\frac{5}{2}} \qquad \beta = 0 \quad \gamma = \sqrt{\frac{5}{2}}$

$[\alpha] = -2 \quad [\beta] = 0 \quad [\gamma] = 1$

Q23. If $y = e^{x\sin(x^3)} + (\tan x)^x$ then $\frac{dy}{dx}$ may be equal to

(A) $e^{x\sin(x^3)}[3x^3\cos(x^3) + \sin(x^3)] + (tanx)^x[ln\tan x + 2x\cos ec2x]$

(B) $e^{x\sin(x^3)}[x^3\cos(x^3) + \sin(x^3)] + (tanx)^x[ln\tan x + 2x\cos ec2x]$

(C) $e^{x\sin(x^3)}[x^3\sin(x^3) + \cos(x^3)] + (tanx)^x[ln\tan x + 2x\cos ec2x]$

(D) $e^{x\sin(x^3)}[3x^3\cos(x^3) + \sin(x^3)] + (\tan x)^x[ln\tan x + \frac{x\sec^2 x}{\tan x}]$

Ans. A, D

Sol. $y = e^{x\ sin\ (x^3)} + \tan(\tan x)^4$

$y = f(x) + g(x)$

$\frac{dy}{dx}\ f'(x) + g'(x)$

$f(x) = e^{x\ sin\ (x^3)}\ \left(3x^3\ cos(x^3)\ + sin(x^3)\right)$

$g'(x) = e^{ln(tan\ x)}\ \left(ln\ (tan\ x) + \frac{x\ sec^2\ x}{tan\ x}\right)$

$= (\tan x)^x\ \left(ln\ (tan\ x)\ + \frac{2x}{sin\ 2x}\right)$

Q24. Let $f(x) = x + (1 - x)x^2 + (1 - x)(1 - x^2)x^3 + ... + (1 - x)(1 - x^2) ..... (1 - x^{n-1})$ xn ; $(n \geq 4)$ then

(A) $f(x) = -\prod_{r=1}^{n}(1 - x^r)$ (B) $f(x) = 1 - \prod_{r=1}^{n}(1 - x^r)$ (C) $f'(x) = (1 - f(x))\left(\sum_{r=1}^{n}\frac{rx^{r-1}}{1 - X^r}\right)$

(D) $f'(x) = f(x))\left(\sum_{r=1}^{n}\frac{rx^{r-1}}{1 - X^r}\right)$

Ans. B, C

Sol. $1 - f(x) = 1 - x\ (1 - x)\ x^2 - (1 - x)(1 - x^2)\ x^3$ ......

$= (1 - x)(1 - x^2) - (1 - x)(1 - x^2)x^3 +$ ....

$1 - f(x)\ (1 - x)(1 - x^2)(1 - x^3)$ .... $(1 - x^n)$

$f(x) = 1 - \prod_{r=1}^{n}(1 - x^r) \quad \Rightarrow$ B is corect

$1 - f(x) = \prod_{r=1}^{n} (1 - x^r)$

Take logarithm

$\log (1 + f(x)) = \sum_{r=1}^{n} log \ (1 - r^r)$

Diff. wrt x

$\frac{-f'(x)}{1 - f(x)} = \sum_{r=1}^{n} \quad \frac{rx^{r-1}}{1 - x^r}$

$f'(x) = (1 - f(x)) \left(\sum_{r=1}^{n} \frac{rx^{r-1}}{1 - x^r} \right) \Rightarrow$ C is correct